

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409604
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Yang; Xi et al.

Systems and methods for additive manufacturing

Abstract

An additive manufacturing apparatus includes a resin support configured to support a first resin and a second resin. A support plate includes a window. A stage is configured to hold one or more cured layers of the resin to form a component positioned opposite the support plate. A radiant energy device is positioned on an opposite side of the resin support from the stage and is operable to generate and project radiant energy in a patterned image through the window. An actuator assembly is configured to move the stage in a Z-axis direction and in a Y-axis direction.

Inventors: Yang; Xi (Mason, OH), Steele; William Joseph (Lawrenceburg, IN), Barnhill; Christopher David (Cincinnati, OH), Dubelman; Meredith Elissa (Liberty Township, OH), Muhlenkamp; Trent William (Cincinnati, OH), Thompson; Mary Kathryn (Fairfield Township, OH)

Applicant: General Electric Company (Schenectady, NY)

Family ID: 85640748

Assignee: General Electric Company (Evendale, OH)

Appl. No.: 18/184814

Filed: March 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230302728 A1	Sep. 28, 2023

Related U.S. Application Data

us-provisional-application US 63322864 20220323

Publication Classification

Int. Cl.: **B29C64/245** (20170101); **B29C64/124** (20170101); **B29C64/236** (20170101); **B29C64/336** (20170101); **B33Y10/00** (20150101); **B33Y30/00** (20150101)

U.S. Cl.:

CPC **B29C64/245** (20170801); **B29C64/124** (20170801); **B29C64/236** (20170801); **B29C64/336** (20170801); **B33Y10/00** (20141201); **B33Y30/00** (20141201);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1990749	12/1934	Phillips et al.	N/A	N/A
2259517	12/1940	Drenkard, Jr.	N/A	N/A
3264103	12/1965	Cohen et al.	N/A	N/A
3395014	12/1967	Cohen et al.	N/A	N/A
3486482	12/1968	Hunger	N/A	N/A
3710846	12/1972	Properzi	N/A	N/A
3875067	12/1974	DeSorgo et al.	N/A	N/A
3991149	12/1975	Hurwitt	N/A	N/A
4041476	12/1976	Swainson	N/A	N/A
4292827	12/1980	Waugh	N/A	N/A
4575330	12/1985	Hull	N/A	N/A
4752498	12/1987	Fudim	N/A	N/A
4945032	12/1989	Murphy et al.	N/A	N/A
5015312	12/1990	Kinzie	N/A	N/A
5026146	12/1990	Hug et al.	N/A	N/A
5031120	12/1990	Pomerantz et al.	N/A	N/A
5058988	12/1990	Spence et al.	N/A	N/A
5059021	12/1990	Spence et al.	N/A	N/A
5088047	12/1991	Bynum	N/A	N/A
5094935	12/1991	Vassiliou et al.	N/A	N/A
5096530	12/1991	Cohen	N/A	N/A
5104592	12/1991	Hull et al.	N/A	N/A
5123734	12/1991	Spence et al.	N/A	N/A
5126259	12/1991	Weiss et al.	N/A	N/A
5126529	12/1991	Weiss et al.	N/A	N/A
5133987	12/1991	Spence et al.	N/A	N/A
5162167	12/1991	Minh et al.	N/A	N/A
5174931	12/1991	Almquist et al.	N/A	N/A
5175077	12/1991	Grossa	N/A	N/A
5182055	12/1992	Allison et al.	N/A	N/A
5192559	12/1992	Hull et al.	N/A	N/A
5203944	12/1992	Prinz et al.	N/A	N/A
5204055	12/1992	Sachs et al.	N/A	N/A
5207371	12/1992	Prinz et al.	N/A	N/A
5236326	12/1992	Grossa	N/A	N/A
5236637	12/1992	Hull	N/A	N/A

5236812	12/1992	Vassiliou et al.	N/A	N/A
5247180	12/1992	Mitcham et al.	N/A	N/A
5248456	12/1992	Evans, Jr. et al.	N/A	N/A
5258146	12/1992	Almquist et al.	N/A	N/A
5314711	12/1993	Baccini	N/A	N/A
5340656	12/1993	Sachs et al.	N/A	N/A
5387380	12/1994	Cima et al.	N/A	N/A
5432045	12/1994	Narukawa et al.	N/A	N/A
5447822	12/1994	Hull et al.	N/A	N/A
5454069	12/1994	Knapp et al.	N/A	N/A
5460758	12/1994	Langer et al.	N/A	N/A
5496682	12/1995	Quadir et al.	N/A	N/A
5610824	12/1996	Vinson et al.	N/A	N/A
5626919	12/1996	Chapman et al.	N/A	N/A
5650260	12/1996	Onishi	N/A	N/A
5660621	12/1996	Bredt	N/A	N/A
5665401	12/1996	Serbin et al.	N/A	N/A
5688464	12/1996	Jacobs et al.	N/A	N/A
5693144	12/1996	Jacobs et al.	N/A	N/A
5697043	12/1996	Baskaran et al.	N/A	N/A
5717599	12/1997	Menhennett et al.	N/A	N/A
5718279	12/1997	Saoth et al.	N/A	N/A
5746833	12/1997	Gerhardt	N/A	N/A
5764521	12/1997	Batchelder et al.	N/A	N/A
5807437	12/1997	Sachs et al.	N/A	N/A
5824184	12/1997	Kamijo et al.	N/A	N/A
5851465	12/1997	Bredt	N/A	N/A
5866058	12/1998	Batchelder et al.	N/A	N/A
5895547	12/1998	Kathrein et al.	N/A	N/A
5900207	12/1998	Danforth et al.	N/A	N/A
5939008	12/1998	Comb et al.	N/A	N/A
5940674	12/1998	Sachs et al.	N/A	N/A
5945058	12/1998	Manners et al.	N/A	N/A
5968561	12/1998	Batchelder et al.	N/A	N/A
5980813	12/1998	Narang et al.	N/A	N/A
5985204	12/1998	Otsuka et al.	N/A	N/A
6051179	12/1999	Hagenau	N/A	N/A
6067480	12/1999	Stuffle et al.	N/A	N/A
6068367	12/1999	Fabbri	N/A	N/A
6110411	12/1999	Clausen et al.	N/A	N/A
6146567	12/1999	Sachs et al.	N/A	N/A
6193923	12/2000	Leyden et al.	N/A	N/A
6200646	12/2000	Neckers et al.	N/A	N/A
6206672	12/2000	Grenda	N/A	N/A
6363606	12/2001	Johnson et al.	N/A	N/A
6375451	12/2001	Robinson et al.	N/A	N/A
6376148	12/2001	Liu et al.	N/A	N/A
6391245	12/2001	Smith	N/A	N/A
6399010	12/2001	Guertin et al.	N/A	N/A
6401002	12/2001	Jang et al.	N/A	N/A
6403002	12/2001	van der Geest	N/A	N/A
6436520	12/2001	Yamamoto	N/A	N/A

6450393	12/2001	Doumanidis et al.	N/A	N/A
6463349	12/2001	White et al.	N/A	N/A
6471800	12/2001	Jang et al.	N/A	N/A
6500378	12/2001	Smith	N/A	N/A
6512869	12/2002	Imayama et al.	N/A	N/A
6543506	12/2002	Phillips	N/A	N/A
6575218	12/2002	Burns et al.	N/A	N/A
6596224	12/2002	Sachs et al.	N/A	N/A
6641897	12/2002	Gervasi	N/A	N/A
6649113	12/2002	Manners et al.	N/A	N/A
6660209	12/2002	Leyden et al.	N/A	N/A
6668892	12/2002	Vasilakes et al.	N/A	N/A
6682598	12/2003	Steinmueller et al.	N/A	N/A
6780368	12/2003	Liu et al.	N/A	N/A
6786711	12/2003	Koch et al.	N/A	N/A
6838035	12/2004	Ederer et al.	N/A	N/A
6850334	12/2004	Gothait	N/A	N/A
6852272	12/2004	Artz et al.	N/A	N/A
6896839	12/2004	Kubo et al.	N/A	N/A
6914406	12/2004	Wilkes et al.	N/A	N/A
6930144	12/2004	Oriakhi	N/A	N/A
6947058	12/2004	Elmqvist	N/A	N/A
6966960	12/2004	Boyd et al.	N/A	N/A
6974521	12/2004	Schermer	N/A	N/A
6986654	12/2005	Imiolek et al.	N/A	N/A
7008209	12/2005	Iskra et al.	N/A	N/A
7016738	12/2005	Karunasiri	N/A	N/A
7022207	12/2005	Hirsch	N/A	N/A
7045738	12/2005	Kovacevic et al.	N/A	N/A
7052263	12/2005	John	N/A	N/A
7070250	12/2005	Lester et al.	N/A	N/A
7074029	12/2005	Stockwell et al.	N/A	N/A
7084875	12/2005	Plante	N/A	N/A
7087109	12/2005	Bredr et al.	N/A	N/A
7158849	12/2006	Huang et al.	N/A	N/A
7164420	12/2006	Ard	N/A	N/A
7195472	12/2006	John	N/A	N/A
7261542	12/2006	Hickerson et al.	N/A	N/A
7270528	12/2006	Sherwood	N/A	N/A
7300613	12/2006	Sano et al.	N/A	N/A
7351304	12/2007	Liang et al.	N/A	N/A
7402219	12/2007	Graf	N/A	N/A
7438846	12/2007	John	N/A	N/A
7455804	12/2007	Patel et al.	N/A	N/A
7520740	12/2008	Wahlstrom et al.	N/A	N/A
7550518	12/2008	Bredt et al.	N/A	N/A
7555726	12/2008	Kurtenbach et al.	N/A	N/A
7569174	12/2008	Ruatta et al.	N/A	N/A
7572403	12/2008	Gu et al.	N/A	N/A
7575682	12/2008	Olsta et al.	N/A	N/A
7578958	12/2008	Patel et al.	N/A	N/A
7614866	12/2008	Sperry et al.	N/A	N/A

7614886	12/2008	Sperry et al.	N/A	N/A
7636610	12/2008	Schillen et al.	N/A	N/A
7698947	12/2009	Sarr	N/A	N/A
7706910	12/2009	Hull et al.	N/A	N/A
7742060	12/2009	Maillot	N/A	N/A
7758799	12/2009	Hull et al.	N/A	N/A
7767132	12/2009	Patel et al.	N/A	N/A
7771183	12/2009	Hull et al.	N/A	N/A
7780429	12/2009	Kikuchi	N/A	N/A
7783371	12/2009	John et al.	N/A	N/A
7785093	12/2009	Holmboe et al.	N/A	N/A
7790093	12/2009	Shkolnik et al.	N/A	N/A
7795349	12/2009	Bredt et al.	N/A	N/A
7815826	12/2009	Serdy et al.	N/A	N/A
7845930	12/2009	Shkolnik et al.	N/A	N/A
7867302	12/2010	Nevoret et al.	N/A	N/A
7892474	12/2010	Shkolnik et al.	N/A	N/A
7894921	12/2010	John et al.	N/A	N/A
7931460	12/2010	Scott et al.	N/A	N/A
7962238	12/2010	Shkolnik et al.	N/A	N/A
7964047	12/2010	Ishida	N/A	N/A
7995073	12/2010	Shemanarev et al.	N/A	N/A
8003040	12/2010	El-Siblani	N/A	N/A
8029642	12/2010	Hagman	N/A	N/A
8048261	12/2010	McCowin	N/A	N/A
8070473	12/2010	Kozlak	N/A	N/A
8071055	12/2010	Newcombe	N/A	N/A
8096262	12/2011	Ederer et al.	N/A	N/A
8105066	12/2011	Sperry et al.	N/A	N/A
8110135	12/2011	El-Siblani	N/A	N/A
8126580	12/2011	El-Siblani et al.	N/A	N/A
8157908	12/2011	Williams	N/A	N/A
8185229	12/2011	Davidson	N/A	N/A
8191500	12/2011	Dohring et al.	N/A	N/A
8211226	12/2011	Bredt et al.	N/A	N/A
8232444	12/2011	Bar Nathan et al.	N/A	N/A
8259103	12/2011	Glueck et al.	N/A	N/A
8269767	12/2011	Glueck et al.	N/A	N/A
8282866	12/2011	Hiraide	N/A	N/A
8326024	12/2011	Shkolnik	N/A	N/A
8372330	12/2012	El-Siblani et al.	N/A	N/A
8394313	12/2012	El-Siblani et al.	N/A	N/A
8413578	12/2012	Doyle	N/A	N/A
8424580	12/2012	Anderson et al.	N/A	N/A
8444903	12/2012	Lyons et al.	N/A	N/A
8454879	12/2012	Kuzusako et al.	N/A	N/A
8475946	12/2012	Dion et al.	N/A	N/A
8506862	12/2012	Giller et al.	N/A	N/A
8506870	12/2012	Hochsmann et al.	N/A	N/A
8513562	12/2012	Bichsel	N/A	N/A
8522159	12/2012	Kurtenbach et al.	N/A	N/A
8540501	12/2012	Yasukochi	N/A	N/A

8568646	12/2012	Wang et al.	N/A	N/A
8568649	12/2012	Balistreri et al.	N/A	N/A
8593083	12/2012	Firhoj et al.	N/A	N/A
8616872	12/2012	Matsui et al.	N/A	N/A
8623264	12/2013	Rohner et al.	N/A	N/A
8636494	12/2013	Gothait et al.	N/A	N/A
8636496	12/2013	Das et al.	N/A	N/A
8658076	12/2013	El-Siblani	N/A	N/A
8663568	12/2013	Bar Nathan et al.	N/A	N/A
8666142	12/2013	Shkolnik et al.	N/A	N/A
8703037	12/2013	Hull et al.	N/A	N/A
8715832	12/2013	Ederer et al.	N/A	N/A
8718522	12/2013	Chillscyzn et al.	N/A	N/A
8737862	12/2013	Manico et al.	N/A	N/A
8741194	12/2013	Ederer et al.	N/A	N/A
8741203	12/2013	Liska et al.	N/A	N/A
8744184	12/2013	Ameline et al.	N/A	N/A
8761918	12/2013	Silverbrook	N/A	N/A
8801418	12/2013	El-Siblani et al.	N/A	N/A
8805064	12/2013	Ameline et al.	N/A	N/A
8815143	12/2013	John et al.	N/A	N/A
8844133	12/2013	Fuller	N/A	N/A
8845316	12/2013	Schillen et al.	N/A	N/A
8845953	12/2013	Balistreri et al.	N/A	N/A
8862260	12/2013	Shkolnik et al.	N/A	N/A
8872024	12/2013	Jamar et al.	N/A	N/A
8873024	12/2013	Jamar et al.	N/A	N/A
8876513	12/2013	Lim et al.	N/A	N/A
8877115	12/2013	Elsey	N/A	N/A
8888480	12/2013	Yoo et al.	N/A	N/A
8915728	12/2013	Mironets et al.	N/A	N/A
8926304	12/2014	Chen	N/A	N/A
8932511	12/2014	Napendensky	N/A	N/A
8968625	12/2014	Tan	N/A	N/A
8974717	12/2014	Maguire et al.	N/A	N/A
8991211	12/2014	Arlotti et al.	N/A	N/A
8992816	12/2014	Jonasson et al.	N/A	N/A
8998601	12/2014	Busato	N/A	N/A
9011982	12/2014	Muller et al.	N/A	N/A
9031680	12/2014	Napadensky	N/A	N/A
9063376	12/2014	Mizumura	N/A	N/A
9064922	12/2014	Nakajima et al.	N/A	N/A
9067359	12/2014	Rohner et al.	N/A	N/A
9067360	12/2014	Wehning et al.	N/A	N/A
9067361	12/2014	El-Siblani	N/A	N/A
9073260	12/2014	El-Siblani et al.	N/A	N/A
9079357	12/2014	Ebert et al.	N/A	N/A
9101321	12/2014	Kiesser	N/A	N/A
9149986	12/2014	Huang et al.	N/A	N/A
9150032	12/2014	Roof et al.	N/A	N/A
9153052	12/2014	Ameline et al.	N/A	N/A
9159155	12/2014	Andersen	N/A	N/A

9186847	12/2014	Fruth et al.	N/A	N/A
9193112	12/2014	Ohkusa et al.	N/A	N/A
9205601	12/2014	DeSimone et al.	N/A	N/A
9211678	12/2014	DeSimone et al.	N/A	N/A
9216546	12/2014	DeSimone et al.	N/A	N/A
9221100	12/2014	Schwarze et al.	N/A	N/A
9233504	12/2015	Douglas et al.	N/A	N/A
9248600	12/2015	Goodman et al.	N/A	N/A
9259880	12/2015	Chen	N/A	N/A
9308690	12/2015	Boyer et al.	N/A	N/A
9327385	12/2015	Webb et al.	N/A	N/A
9346217	12/2015	Huang et al.	N/A	N/A
9346218	12/2015	Chen et al.	N/A	N/A
9360757	12/2015	DeSimone et al.	N/A	N/A
9364848	12/2015	Silverbrook	N/A	N/A
9403322	12/2015	Das et al.	N/A	N/A
9403324	12/2015	Ederer et al.	N/A	N/A
9415443	12/2015	Ljungblad et al.	N/A	N/A
9415544	12/2015	Kerekes et al.	N/A	N/A
9415547	12/2015	Chen et al.	N/A	N/A
9429104	12/2015	Fuller	N/A	N/A
9434107	12/2015	Zenere	N/A	N/A
9446557	12/2015	Zenere et al.	N/A	N/A
9453142	12/2015	Rolland et al.	N/A	N/A
9456884	12/2015	Uckelmann et al.	N/A	N/A
9457374	12/2015	Hibbs et al.	N/A	N/A
9463488	12/2015	Ederer et al.	N/A	N/A
9469074	12/2015	Ederer et al.	N/A	N/A
9486944	12/2015	El-Siblani et al.	N/A	N/A
9486964	12/2015	Joyce	N/A	N/A
9487443	12/2015	Watanabe	N/A	N/A
9498920	12/2015	DeSimone et al.	N/A	N/A
9498921	12/2015	Teulet	N/A	N/A
9511546	12/2015	Chen et al.	N/A	N/A
9517591	12/2015	Yoo et al.	N/A	N/A
9517592	12/2015	Yoo et al.	N/A	N/A
9527244	12/2015	El-Siblani	N/A	N/A
9527272	12/2015	Steele	N/A	N/A
9529371	12/2015	Nakamura	N/A	N/A
9533450	12/2016	El-Siblani et al.	N/A	N/A
9539762	12/2016	Durand et al.	N/A	N/A
9545753	12/2016	Costabeber	N/A	N/A
9545784	12/2016	Nakamura	N/A	N/A
9550326	12/2016	Costabeber	N/A	N/A
9561622	12/2016	Das et al.	N/A	N/A
9561623	12/2016	El-Siblani et al.	N/A	N/A
9578695	12/2016	Jerby et al.	N/A	N/A
9579852	12/2016	Okamoto	N/A	N/A
9581530	12/2016	Guthrie et al.	N/A	N/A
9592635	12/2016	Ebert et al.	N/A	N/A
9604411	12/2016	Rogren	N/A	N/A
9610616	12/2016	Chen et al.	N/A	N/A

9616620	12/2016	Hoechsmann et al.	N/A	N/A
9632037	12/2016	Chen et al.	N/A	N/A
9632420	12/2016	Allanic	N/A	N/A
9632983	12/2016	Ueda et al.	N/A	N/A
9636873	12/2016	Joyce	N/A	N/A
9649812	12/2016	Hartmann et al.	N/A	N/A
9649815	12/2016	Atwood et al.	N/A	N/A
9656344	12/2016	Kironn et al.	N/A	N/A
9670371	12/2016	Pervan et al.	N/A	N/A
9676143	12/2016	Kashani-Shirazi	N/A	N/A
9676963	12/2016	Rolland et al.	N/A	N/A
9682166	12/2016	Watanabe	N/A	N/A
9682425	12/2016	Xu et al.	N/A	N/A
9688027	12/2016	Batchelder et al.	N/A	N/A
9707720	12/2016	Chen et al.	N/A	N/A
9720363	12/2016	Chillscyzn et al.	N/A	N/A
9738034	12/2016	Gruber et al.	N/A	N/A
9738564	12/2016	Capobianco et al.	N/A	N/A
9751292	12/2016	Jamar et al.	N/A	N/A
9764513	12/2016	Stampfl et al.	N/A	N/A
9764535	12/2016	Xie et al.	N/A	N/A
9821546	12/2016	Schaafsma et al.	N/A	N/A
9862146	12/2017	Driessen et al.	N/A	N/A
9862150	12/2017	Chen et al.	N/A	N/A
9868255	12/2017	Comb et al.	N/A	N/A
9885987	12/2017	Chillscysn et al.	N/A	N/A
9895843	12/2017	Lobovsky et al.	N/A	N/A
9901983	12/2017	Hovel et al.	N/A	N/A
9908293	12/2017	Yoo et al.	N/A	N/A
9919474	12/2017	Napadensky	N/A	N/A
9919515	12/2017	Daniell et al.	N/A	N/A
9950368	12/2017	Lampenscherf et al.	N/A	N/A
9956727	12/2017	Steele	N/A	N/A
9962767	12/2017	Buller et al.	N/A	N/A
9981411	12/2017	Green et al.	N/A	N/A
10000023	12/2017	El-Siblani et al.	N/A	N/A
10011076	12/2017	El-Siblani et al.	N/A	N/A
10061302	12/2017	Jacobs et al.	N/A	N/A
10071422	12/2017	Buller et al.	N/A	N/A
10124532	12/2017	El-Siblani et al.	N/A	N/A
10150254	12/2017	Bauman et al.	N/A	N/A
10155345	12/2017	Ermoshkin et al.	N/A	N/A
10155882	12/2017	Rolland et al.	N/A	N/A
10183330	12/2018	Buller et al.	N/A	N/A
10183444	12/2018	Campbell	N/A	N/A
10240066	12/2018	Rolland et al.	N/A	N/A
10245784	12/2018	Teken et al.	N/A	N/A
10317882	12/2018	de Pena et al.	N/A	N/A
10336055	12/2018	Das et al.	N/A	N/A
10336057	12/2018	Moore et al.	N/A	N/A
10350823	12/2018	Rolland et al.	N/A	N/A
10357956	12/2018	Usami et al.	N/A	N/A

10406748	12/2018	Honda	N/A	N/A
10612112	12/2019	Yang et al.	N/A	N/A
10639843	12/2019	Yuan et al.	N/A	N/A
10682808	12/2019	Fujita et al.	N/A	N/A
10695988	12/2019	Hanyu et al.	N/A	N/A
10717212	12/2019	Parkinson et al.	N/A	N/A
10737479	12/2019	El-Siblani et al.	N/A	N/A
2002/0164069	12/2001	Nagano et al.	N/A	N/A
2003/0180171	12/2002	Artz et al.	N/A	N/A
2003/0209836	12/2002	Sherwood	N/A	N/A
2005/0012239	12/2004	Nakashima	N/A	N/A
2005/0019016	12/2004	Nakashika et al.	N/A	N/A
2006/0230984	12/2005	Bredt et al.	N/A	N/A
2006/0248062	12/2005	Libes et al.	N/A	N/A
2007/0063366	12/2006	Cunningham et al.	N/A	N/A
2007/0116937	12/2006	Lazzerini	N/A	N/A
2008/0170112	12/2007	Hull et al.	N/A	N/A
2008/0224352	12/2007	Narukawa et al.	N/A	N/A
2008/0241404	12/2007	Allaman et al.	N/A	N/A
2010/0003619	12/2009	Das et al.	N/A	N/A
2010/0196694	12/2009	Yamazaki et al.	N/A	N/A
2010/0290016	12/2009	Kaehr et al.	N/A	N/A
2011/0089610	12/2010	El-Siblani et al.	N/A	N/A
2011/0101570	12/2010	John et al.	N/A	N/A
2011/0162989	12/2010	Ducker et al.	N/A	N/A
2011/0207057	12/2010	Hull et al.	N/A	N/A
2012/0195994	12/2011	El-Siblani et al.	N/A	N/A
2012/0292800	12/2011	Higuchi et al.	N/A	N/A
2013/0008879	12/2012	Bichsel	N/A	N/A
2013/0140741	12/2012	El-Siblani et al.	N/A	N/A
2014/0099476	12/2013	Subramanian et al.	N/A	N/A
2014/0103581	12/2013	Das et al.	N/A	N/A
2014/0200865	12/2013	Lehmann et al.	N/A	N/A
2014/0239554	12/2013	El-Siblani et al.	N/A	N/A
2014/0275317	12/2013	Moussa	N/A	N/A
2014/0319735	12/2013	El-Siblani et al.	N/A	N/A
2014/0322374	12/2013	El-Siblani et al.	N/A	N/A
2014/0332507	12/2013	Fockele	N/A	N/A
2014/0339741	12/2013	Aghababaie et al.	N/A	N/A
2014/0348691	12/2013	Ljungblad et al.	N/A	N/A
2014/0348692	12/2013	Bessac et al.	N/A	N/A
2015/0004042	12/2014	Nimal	N/A	N/A
2015/0004046	12/2014	Graham et al.	N/A	N/A
2015/0056365	12/2014	Miyoshi	N/A	N/A
2015/0086409	12/2014	Hellestam	N/A	N/A
2015/0102531	12/2014	El-Siblani et al.	N/A	N/A
2015/0104563	12/2014	Lowe et al.	N/A	N/A
2015/0140152	12/2014	Chen	N/A	N/A
2015/0140155	12/2014	Ohno et al.	N/A	N/A
2015/0145174	12/2014	Comb	N/A	N/A
2015/0158111	12/2014	Schwarze et al.	N/A	N/A
2015/0165695	12/2014	Chen et al.	N/A	N/A

2015/0210013	12/2014	Teulet	N/A	N/A
2015/0224710	12/2014	El-Siblani	N/A	N/A
2015/0231828	12/2014	El-Siblani et al.	N/A	N/A
2015/0231831	12/2014	El-Siblani	N/A	N/A
2015/0246487	12/2014	El-Siblani	N/A	N/A
2015/0251351	12/2014	Feygin	N/A	N/A
2015/0268099	12/2014	Craig et al.	N/A	N/A
2015/0298396	12/2014	Chen et al.	N/A	N/A
2015/0301517	12/2014	Chen et al.	N/A	N/A
2015/0306819	12/2014	Ljungblad	N/A	N/A
2015/0306825	12/2014	Chen et al.	N/A	N/A
2015/0321421	12/2014	Ding	N/A	N/A
2015/0352668	12/2014	Scott et al.	N/A	N/A
2015/0352791	12/2014	Chen et al.	N/A	N/A
2015/0355553	12/2014	Allanic	N/A	N/A
2015/0375452	12/2014	Huang et al.	N/A	N/A
2016/0016361	12/2015	Lobovsky et al.	N/A	N/A
2016/0031010	12/2015	O'Neill et al.	N/A	N/A
2016/0046075	12/2015	DeSimone et al.	N/A	N/A
2016/0046080	12/2015	Thomas et al.	N/A	N/A
2016/0052205	12/2015	FrantzDale	N/A	N/A
2016/0059484	12/2015	DeSimone et al.	N/A	N/A
2016/0059485	12/2015	Ding et al.	N/A	N/A
2016/0067921	12/2015	Willis et al.	N/A	N/A
2016/0082662	12/2015	Majer	N/A	N/A
2016/0082671	12/2015	Joyce	N/A	N/A
2016/0096332	12/2015	Chen et al.	N/A	N/A
2016/0107340	12/2015	Joyce	N/A	N/A
2016/0107383	12/2015	Dikovsky et al.	N/A	N/A
2016/0107387	12/2015	Ooba et al.	N/A	N/A
2016/0129631	12/2015	Chen et al.	N/A	N/A
2016/0137839	12/2015	Rolland et al.	N/A	N/A
2016/0167160	12/2015	Hellestam	N/A	N/A
2016/0176114	12/2015	Tsai et al.	N/A	N/A
2016/0184931	12/2015	Green	N/A	N/A
2016/0193785	12/2015	Bell et al.	N/A	N/A
2016/0214327	12/2015	Ucklemann et al.	N/A	N/A
2016/0221262	12/2015	Das et al.	N/A	N/A
2016/0243649	12/2015	Zheng et al.	N/A	N/A
2016/0303798	12/2015	Mironets et al.	N/A	N/A
2016/0332386	12/2015	Kuijpers	N/A	N/A
2016/0361871	12/2015	Jeng et al.	N/A	N/A
2016/0361872	12/2015	El-Siblani	N/A	N/A
2017/0008234	12/2016	Cullen et al.	N/A	N/A
2017/0008236	12/2016	Easter et al.	N/A	N/A
2017/0021562	12/2016	El-Siblani et al.	N/A	N/A
2017/0066185	12/2016	Ermoshkin et al.	N/A	N/A
2017/0066196	12/2016	Beard et al.	N/A	N/A
2017/0072635	12/2016	El-Siblani et al.	N/A	N/A
2017/0080641	12/2016	El-Siblani	N/A	N/A
2017/0087670	12/2016	Kalentic et al.	N/A	N/A
2017/0100895	12/2016	Chou et al.	N/A	N/A

2017/0100897	12/2016	Chou et al.	N/A	N/A
2017/0100899	12/2016	El-Siblani et al.	N/A	N/A
2017/0102679	12/2016	Greene et al.	N/A	N/A
2017/0113409	12/2016	Patrov	N/A	N/A
2017/0120332	12/2016	DeMuth et al.	N/A	N/A
2017/0120333	12/2016	DeMuth et al.	N/A	N/A
2017/0120334	12/2016	DeMuth et al.	N/A	N/A
2017/0120335	12/2016	DeMuth et al.	N/A	N/A
2017/0120336	12/2016	DeMuth et al.	N/A	N/A
2017/0120387	12/2016	DeMuth et al.	N/A	N/A
2017/0120518	12/2016	DeMuth et al.	N/A	N/A
2017/0120529	12/2016	DeMuth et al.	N/A	N/A
2017/0120530	12/2016	DeMuth et al.	N/A	N/A
2017/0120537	12/2016	DeMuth et al.	N/A	N/A
2017/0120538	12/2016	DeMuth et al.	N/A	N/A
2017/0123222	12/2016	DeMuth et al.	N/A	N/A
2017/0123237	12/2016	DeMuth et al.	N/A	N/A
2017/0136688	12/2016	Knecht et al.	N/A	N/A
2017/0136708	12/2016	Das et al.	N/A	N/A
2017/0157841	12/2016	Green	N/A	N/A
2017/0157862	12/2016	Bauer	N/A	N/A
2017/0165916	12/2016	El-Siblani	N/A	N/A
2017/0173865	12/2016	Dikovsky et al.	N/A	N/A
2017/0182708	12/2016	Lin et al.	N/A	N/A
2017/0190120	12/2016	Bloome et al.	N/A	N/A
2017/0276651	12/2016	Hall	N/A	N/A
2017/0284971	12/2016	Hall	N/A	N/A
2017/0291804	12/2016	Craft et al.	N/A	N/A
2017/0297108	12/2016	Gibson et al.	N/A	N/A
2017/0297109	12/2016	Gibson et al.	N/A	N/A
2017/0305136	12/2016	Elsey	N/A	N/A
2017/0326786	12/2016	Yuan et al.	N/A	N/A
2017/0326807	12/2016	Greene et al.	N/A	N/A
2017/0368816	12/2016	Batchelder et al.	N/A	N/A
2018/0001567	12/2017	Juan et al.	N/A	N/A
2018/0015672	12/2017	Shusteff et al.	N/A	N/A
2018/0043619	12/2017	Kim et al.	N/A	N/A
2018/0056585	12/2017	Du Toit	N/A	N/A
2018/0056604	12/2017	Sands et al.	N/A	N/A
2018/0079137	12/2017	Herzog et al.	N/A	N/A
2018/0085998	12/2017	von Burg	N/A	N/A
2018/0117790	12/2017	Yun	N/A	N/A
2018/0169969	12/2017	Deleon et al.	N/A	N/A
2018/0200948	12/2017	Kuijpers et al.	N/A	N/A
2018/0201021	12/2017	Beaver et al.	N/A	N/A
2018/0229332	12/2017	Tsai et al.	N/A	N/A
2018/0229436	12/2017	Gu et al.	N/A	N/A
2018/0272603	12/2017	MacCormack et al.	N/A	N/A
2018/0272608	12/2017	Yun	N/A	N/A
2018/0345600	12/2017	Holford et al.	N/A	N/A
2018/0370214	12/2017	Comb et al.	N/A	N/A
2019/0022937	12/2018	Stelter et al.	N/A	N/A

2019/0039299	12/2018	Busbee et al.	N/A	N/A
2019/0047211	12/2018	Herring et al.	N/A	N/A
2019/0061230	12/2018	Ermoshkin et al.	N/A	N/A
2019/0112499	12/2018	Rolland et al.	N/A	N/A
2019/0126548	12/2018	Barnhart et al.	N/A	N/A
2019/0232550	12/2018	Mark et al.	N/A	N/A
2019/0240932	12/2018	Graf	N/A	N/A
2019/0263054	12/2018	Kotler et al.	N/A	N/A
2019/0283316	12/2018	Rolland et al.	N/A	N/A
2019/0344381	12/2018	Pomerantz et al.	N/A	N/A
2019/0389137	12/2018	Frohnmaier et al.	N/A	N/A
2020/0001398	12/2019	Mellor et al.	N/A	N/A
2020/0079008	12/2019	Chowdry et al.	N/A	N/A
2020/0079017	12/2019	MacNeish, III et al.	N/A	N/A
2020/0108553	12/2019	Rogren	N/A	N/A
2020/0164437	12/2019	Goth et al.	N/A	N/A
2020/0198224	12/2019	Dubelman	N/A	B29C 64/245
2020/0230938	12/2019	Menchik et al.	N/A	N/A
2020/0247040	12/2019	Green	N/A	N/A
2020/0290275	12/2019	Dubelman et al.	N/A	N/A
2020/0307075	12/2019	Mattes et al.	N/A	N/A
2020/0376775	12/2019	Das et al.	N/A	N/A
2021/0046695	12/2020	Thompson et al.	N/A	N/A
2021/0276248	12/2020	Boehm et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101628477	12/2009	CN	N/A
103210344	12/2012	CN	N/A
103522546	12/2013	CN	N/A
104175559	12/2013	CN	N/A
104647752	12/2014	CN	N/A
105711101	12/2015	CN	N/A
105773962	12/2015	CN	N/A
107322930	12/2016	CN	N/A
208946717	12/2018	CN	N/A
109968661	12/2018	CN	N/A
111497231	12/2019	CN	N/A
102007010624	12/2007	DE	N/A
448459	12/1990	EP	N/A
557051	12/1992	EP	N/A
1454831	12/2003	EP	N/A
1852244	12/2006	EP	N/A
1864785	12/2006	EP	N/A
1946908	12/2007	EP	N/A
2521524	12/2011	EP	N/A
3053729	12/2015	EP	N/A
3453521	12/2018	EP	N/A
3356121	12/2019	EP	N/A
2311960	12/1996	GB	N/A
H06246839	12/1993	JP	N/A
H09277384	12/1996	JP	N/A
2002370286	12/2001	JP	N/A

2003039564	12/2002	JP	N/A
2004/257929	12/2003	JP	N/A
2016196098	12/2015	JP	N/A
20170108729	12/2016	KR	N/A
102109664	12/2019	KR	N/A
WO9600422	12/1995	WO	N/A
WO9806560	12/1997	WO	N/A
WO0100390	12/2000	WO	N/A
WO2006/077665	12/2005	WO	N/A
WO2006109355	12/2005	WO	N/A
WO2017009368	12/2016	WO	N/A
WO2017098968	12/2016	WO	N/A
WO2017100538	12/2016	WO	N/A
WO2019/159936	12/2018	WO	N/A
WO2020033607	12/2019	WO	N/A
WO2020185553	12/2019	WO	N/A

OTHER PUBLICATIONS

Admatec, Admaflex 300 DLP 3D Printer, Specifications, Features, Design and Functions, Netherlands, 2 Pages. Retrieved Nov. 5, 2020 from Webpage: <https://admateceurope.com/files/10fla369c2239943e6506f27ba920bd4dd9359078e744369695ab6fffbde75c6c?filename=Admaflex%20300%20brochure.pdf&sig=hQyDlzXkSmFOZwjM>. cited by applicant

Carbon, Carbon SpeedCell: Additive Manufacturing Reinvented, Redwood City California, Mar. 16, 2017, 4 Pages. Retrieved from Webpage: <https://www.carbon3d.com/news/carbon-speedcell-additive-manufacturing-reinvented/>. cited by applicant

Carbon, The 3D Printer for Products that Outperform, 8 Pages. Retrieved from Webpage: <https://www.carbon3d.com>. cited by applicant

DDM Systems, Disruptive Technologies for Additive Manufacturing, 2014. Retrieved on Jul. 7, 2020 from Web Link: <http://www.ddmsys.com/>. cited by applicant

Designing Buildings Wiki, Types of Brick Bonding, 6 Pages. Retrieved Mar. 25, 2021 from Webpage: https://www.designingbuildings.co.uk/wiki/Types_of_brick_bonding. cited by applicant

Doctor Blade with Micrometer Screw Gauge, The Tape Casting Warehouse, Inc., Morrisville PA, 6 Pages. Retrieved Mar. 23, 2021 from Webpage: <https://www.drblade.com/>. cited by applicant

Envisiontec, Advanced DLP for Superior 3D Printing, Mar. 9, 2017, 8 Pages. <https://envisiontec.com/wp-content/uploads/2016/12/Why-EnvisionTEC-DLP-3D-Printing-is-Better-rebranded.pdf>. cited by applicant

Feng et al., Exposure Reciprocity Law in Photopolymerization of Multi-Functional Acrylates and Methacrylates, Macromolecular Chemistry and Physics, vol. 208, 2007, pp. 295-306. cited by applicant

Formlabs, An Introduction to Post-Curing SLA 3D Prints, 8 Pages. Retrieved from Webpage: <https://formlabs.com/blog/introduction-post-curing-sla-3d-prints>. cited by applicant

Formlabs, Form Wash & Form Cure, 8 Pages. Retrieved from Webpage: <https://formlabs.com/tools/wash-cure/>. cited by applicant

Hafkamp et al., A Feasibility Study on Process Monitoring and Control in Vat Photopolymerization of Ceramics, Mechatronics, vol. 56, The Netherlands, Dec. 2018, pp. 220-241. Retrieved from <https://doi.org/10.1016/j.mechatronics.2018.02.006>. cited by applicant

KUDO3D, Post-Process Your SLA Prints in 4 Easy Steps, 8 Pages. Retrieved from Webpage: <https://www.kudo3d.com/post-process-your-sla-prints-in-4-easy-steps/>. cited by applicant

Leap, Low-Frequency Sonic Mixing Technology, Energy Efficiency & Renewable Energy, Energy.Gov, 5 Pages. Retrieved Mar. 17, 2021 from Webpage: <https://www.energy.gov/eere/amo/low-frequency-sonic-mixing-technology>. cited by applicant

Lee et al., Development of a 3D Printer Using Scanning Projection Stereolithography, Scientific Reports, vol. 5, Article No. 9875, 2015, 5 pages. <https://www.nature.com/articles/srep09875#s1>. cited by applicant

Lee et al., Large-Area Compatible Laser Sintering Schemes with a Spatially Extended Focused Beam,

Journal, Micromachines, vol. 8, No. 153, Seoul University, Seoul Korea, May 11, 2017, 8 Pages.
<http://dx.doi.org/10.3390/mi8050153>. cited by applicant

Limaye, Multi-Objective Process Planning Method for Mask Projection Stereolithography, Dissertation Georgia Institute of Technology, Dec. 2007, 324 Pages. cited by applicant

Lithoz, 2 Pages. Retrieved from Webpage: <http://www.lithoz.com/en/our-products/cleaning-station>. cited by applicant

Matthews et al., Diode-Based Additive Manufacturing of Metals Using an Optically-Addressable Light Valve, Optic Express Research Article, vol. 25, No. 10, Lawrence Livermore National Laboratory, Livermore CA, May 10, 2017. cited by applicant

Micron3d, Cleaning of Printed Models, YouTube, Dec. 5, 2016, 1 Page. Retrieved from Webpage: <https://www.youtube.com/watch?v=soA1rSsliBY>. cited by applicant

Nussbaum et al., Evaluation of Processing Variables in Large Area Polymer Sintering of Single Layer Components, Solid Freeform Fabrication 2016: Proceedings of the 27.SUP.th .Annual International Solid Freeform Fabrication Symposium—An Additive Manufacturing Conference Reviewed Paper, University of South Florida, Tampa Florida. cited by applicant

Omegasonics, Ultrasonic Cleaning of 3D Printer Parts, YouTube, Feb. 26, 2014, 1 Page. Retrieved from Webpage: <https://www.youtube.com/watch?v=Gxj47OS5ohk>. cited by applicant

Park et al., Development of Multi-Material DLP 3D Printer, Journal of the Korean Society of Manufacturing Technology Engineers, vol. 26, Issue 1, Seoul Korea, Feb. 15, 2017, pp. 100-107.
<https://doi.org/10.7735/ksmte.2017.26.1.100>. cited by applicant

Prodways Tech, Prodways Movinglight Technology Retrieved on Jul. 2, 2020 from Web Link: <https://www.prodways.com/en/the-prodways-movinglight-technology/>. cited by applicant

RAMCO Equipment Corporation, RAMCO RamTough-Fully Automated Wash/Rinse/Dry System, YouTube, Jul. 9, 2013, 1 Page. Retrieved from Webpage: <https://www.youtube.com/watch?v=i8S5Oc3FVFU>. cited by applicant

Ricoh Imaging Company Ltd., The Advanced Pixel Shift Resolution System II for Super-High-Resolution Images, Pentax K-1 Mark II, Pixel Shift Resolution System, 4 Pages. Retrieved on Mar. 30, 2021 from Webpage: <http://www.ricoh-imaging.co.jp/english/products/k-1-2/feature/02.html>. cited by applicant

Sonics & Materials, Inc., Ultrasonic Food Cutting Equipment, Sonics & Materials, Inc., Retrieved on Jun. 26, 2020, 4 Pages. <https://www.sonics.com/food-cutting>. cited by applicant

Stemmer Imaging, Ultra-High Resolution for Industrial Imaging, Germany, 9 Pages. Retrieved on Mar. 30, 2021 from Webpage: <https://www.stemmer-imaging.com/en/knowledge-base/pixel-shift-technology/>. cited by applicant

Stevenson, Admatec's Ceramic 3D Printers, Ceramic, Metal, Fabbaloo 3D Printing News, Jan. 21, 2019, 8 Pages. Retrieved Nov. 24, 2020 from Weblink: <https://www.fabbaloo.com/blog/2019/1/21/admatecs-ceramic-3d-printers>. cited by applicant

Techmetals, Electroless Nickel (TM 117C), Engineered Metal Finishing & Performance Coatings, 1 Page. Retrieved from Webpage: https://techmetals.com/pdfs/TM_117C.pdf <https://techmetals.com/tm117c-2/>. cited by applicant

Telsonic Ultrasonics, Cutting Awning Fabrics and Sealing the Edge, The Powerhouse of Ultrasonics, 2017, 1 Page. https://www.telsonic.com/fileadmin/applications/AS_206_Cut_Seal_Markisengewebe_EN.pdf. cited by applicant

Telsonic Ultrasonics, Integrated Power Actuator—IPA 3505, Telsonic Ultrasonics, Retrieved Jun. 26, 2020, 2 Pages. <https://www.telsonic.com/en/products/integrated-power-actuator-ipa-3505/>. cited by applicant

Tok et al., Tape Casting of High Dielectric Ceramic Substrates for Microelectronics Packaging, Journal of Materials Engineering and Performance, vol. 8, 1999, pp. 469-472. (Abstract Only)
<https://link.springer.com/article/10.1361/105994999770346783>. cited by applicant

Wikipedia, Pixel Shifting, 2 Pages. Retrieved Mar. 30, 2021 from Webpage: https://en.wikipedia.org/wiki/Pixel_shifting. cited by applicant

Wikipedia, Standing Wave, 11 Pages. Retrieved Mar. 17, 2021 from Webpage: https://en.wikipedia.org/wiki/Standing_wave. cited by applicant

Primary Examiner: Tsui; Yung-Sheng M

Assistant Examiner: Tsui; Melody

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/322,864, filed on Mar. 23, 2022, the contents of which are hereby incorporated by reference in their entirety.

FIELD

(1) The present subject matter relates generally to an additive manufacturing apparatus, and more particularly to assemblies for altering positions of various components of the additive manufacturing apparatus.

BACKGROUND

(2) Additive manufacturing is a process in which material is built up layer-by-layer to form a component. Stereolithography (SLA) is a type of additive manufacturing process, which employs a tank of radiant-energy curable photopolymer “resin” and a curing energy source such as a laser. Similarly, Digital Light Processing (DLP) three-dimensional (3D) printing employs a two-dimensional image projector to build components one layer at a time. For each layer, the energy source draws or flashes a radiation image of the cross section of the component onto the surface of the resin. Exposure to the radiation cures and solidifies the pattern in the resin and joins it to a previously cured layer.

(3) In some instances, additive manufacturing may be accomplished through a “tape casting” process. In this process, a resin is deposited onto a flexible radiotransparent resin support, such as a tape or foil, that is fed out from a supply reel to a build zone. Radiant energy is produced from a radiant energy device and directed through a window to cure the resin to a component that is supported by a stage in the build zone. Once the curing of the first layer is complete, the stage and the resin support are separated from one another. The resin support is then advanced and fresh resin is provided to the build zone. In turn, the first layer of the cured resin is placed onto the fresh resin and cured through the energy device to form an additional layer of the component. Subsequent layers are added to each previous layer until the component is completed. The tape casting process may be used to form various components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

(2) FIG. 1A is a schematic side view of an additive manufacturing apparatus in accordance with various aspects of the present disclosure;

(3) FIG. 1B is a schematic side view of an additive manufacturing apparatus in accordance with various aspects of the present disclosure;

(4) FIG. 2 is a front perspective view of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;

(5) FIG. 3 is a side perspective view of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;

(6) FIG. 4 is a side perspective view of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;

- (7) FIG. 5 is a method of operating the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (8) FIGS. 6-24 illustrate schematic views of the additive manufacturing apparatus during operation of the apparatus in accordance with various aspects of the present disclosure;
- (9) FIG. 25 is a side perspective view of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (10) FIG. 26 is a side perspective view of a frame of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (11) FIG. 27 is a rear perspective view of a frame of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (12) FIG. 28 is a side perspective view of a print head of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (13) FIG. 29 is a cross-sectional view of the print head of the additive manufacturing apparatus taken along the line XXIX-XXIX of FIG. 25 in accordance with various aspects of the present disclosure;
- (14) FIG. 30 is a top plan view of a first slide assembly of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (15) FIG. 31 is a top plan view of a second slide assembly of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (16) FIG. 32 is a side perspective view of the additive manufacturing apparatus with the stage in a first position and the radiant energy device in a first projection position in accordance with various aspects of the present disclosure;
- (17) FIG. 33 is a side perspective view of the additive manufacturing apparatus with the stage in the first position and the radiant energy device in a second projection position in accordance with various aspects of the present disclosure;
- (18) FIG. 34 is a side perspective view of the additive manufacturing apparatus with the stage in a second position and the radiant energy device in a third projection position in accordance with various aspects of the present disclosure;
- (19) FIG. 35 is a side perspective view of the additive manufacturing apparatus with the stage in the second position and the radiant energy device in a fourth projection position in accordance with various aspects of the present disclosure;
- (20) FIG. 36 is a side perspective view of the additive manufacturing apparatus in accordance with various aspects of the present disclosure;
- (21) FIG. 37 is a method of operating the additive manufacturing apparatus in accordance with various aspects of the present disclosure; and
- (22) FIG. 38 depicts an exemplary computing system for an additive manufacturing apparatus in accordance with various aspects of the present disclosure.
- (23) Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present disclosure.

DETAILED DESCRIPTION

- (24) Reference will now be made in detail to present embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the invention.
- (25) As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify a location or importance of the individual components. The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein. The terms “upstream” and “downstream” refer to the relative direction with respect to a resin support movement along the manufacturing apparatus. For example, “upstream” refers to the direction from which the resin support moves, and “downstream” refers to the direction to which the resin support moves. The term “selectively” refers to a component's ability to operate in various states (e.g., an ON state and an OFF state) based on manual

and/or automatic control of the component.

(26) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(27) Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” “generally,” and “substantially,” is not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or apparatus for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a ten percent margin.

(28) Moreover, the technology of the present application will be described in relation to exemplary embodiments. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

(29) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(30) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items can be employed. For example, if a composition or assembly is described as containing components A, B, and/or C, the composition or assembly can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(31) The present disclosure is generally directed to an additive manufacturing apparatus that implements various manufacturing processes such that successive layers of material(s) are provided on each other to “build-up,” layer-by-layer, a three-dimensional component. The successive layers generally cure together to form a monolithic component which may have a variety of integral sub-components. Although additive manufacturing technology is described herein as enabling the fabrication of complex objects by building objects point-by-point, layer-by-layer, variations of the described additive manufacturing apparatus and technology are possible and within the scope of the present subject matter.

(32) The additive manufacturing apparatus can include a support plate, a window supported by the support plate, and a stage moveable relative to the window. The additive manufacturing apparatus can further include a first resin and a second resin that are each deposited as layers having a desired thickness onto a resin support (such as a foil, tape, vat, plate, etc.) that is fed out from a supply reel or a pair of supply reels in an X-axis direction. In various instances, the first resin may be laterally offset from the second resin in a Y-axis direction.

(33) A stage lowers onto the resin such that a working surface defined by one of a surface of the stage or a surface of the work in process component is positioned such that the working surface either is just touching the resin or compressing it between the resin support and the stage and defining a layer thickness. Radiant energy is used to cure the resin through the resin support. Once the curing of the first layer is complete, the stage is retracted, taking the cured material with the stage. The resin support is then advanced to expose a fresh clean section, ready for additional resin to be deposited in a subsequent, new cycle.

(34) In some instances, the additive manufacturing apparatus further includes an actuator assembly including a first actuator configured to move the stage in a Z-axis direction and a second actuator configured to move the stage in a Y-axis direction. The movement of the stage in the Y-axis direction may allow for layers of the component to selectively be formed from the first resin and/or the second resin. As such, a component may be formed from one or more resins.

(35) Referring to the drawings wherein identical reference numerals denote the similar elements

throughout the various views, FIGS. 1A and 1B schematically illustrate an example of one type of suitable apparatus **10** for forming a component **12** created through one or more layers of at least one cured resin R. The apparatus **10** can include one or more of a support plate **14**, a window **16**, a stage **18** that is movable relative to the window **16**, and a radiant energy device **20**, which, in combination, may be used to form any number (e.g., one or more) of additively manufactured components **12**.

(36) In the illustrated example of FIG. 1A, the apparatus **10** includes a feed module **22**, which may include a first roller **22A**, and a take-up module **24**, which may include a second roller **24A**, that are spaced-apart with a resin support **26** extending therebetween. A portion of the resin support **26** can be supported from underneath by the support plate **14**. Suitable mechanical supports (frames, brackets, etc.) and/or alignment devices may be provided for the rollers **22A**, **24A** and the support plate **14**. The first roller **22A** and/or the second roller **24A** can be configured to control the speed and direction of the resin support **26** such that the desired tension and speed is maintained in the resin support **26** through a drive system **28**. By way of example and not limitation, the drive system **28** can be configured as individual motors associated with the first roller **22A** and/or the second roller **24A**. Moreover, various components, such as motors, actuators, feedback sensors, and/or controls can be provided for driving the rollers **22A**, **24A** in such a manner to maintain the resin support **26** tensioned between the aligned rollers **22A**, **24A** and to wind the resin support **26** from the first roller **22A** to the second roller **24A**.

(37) In various embodiments, the window **16** is transparent and can be operably supported by the support plate **14**. Further, the window **16** and the support plate **14** can be integrally formed such that one or more windows **16** are integrated within the support plate **14**. Likewise, the resin support **26** is also transparent or includes portions that are transparent. As used herein, the terms “transparent” and “radiotransparent” refer to a material that allows at least a portion of radiant energy of a selected wavelength to pass through. For example, the radiant energy that passes through the window **16** and the resin support **26** can be in the ultraviolet spectrum, the infrared spectrum, the visible spectrum, or any other practicable radiant energy. Non-limiting examples of transparent materials include polymers, glass, and crystalline minerals, such as sapphire or quartz.

(38) The resin support **26** extends between the feed module **22** and the take-up module **24** and defines a “build surface” **30**, which is shown as being planar, but could alternatively be arcuate (depending on the shape of the support plate **14**). In some instances, the build surface **30** may be defined by the resin support **26** and be positioned to face the stage **18** with the window **16** on an opposing side of the resin support **26** from the stage **18**. For purposes of convenient description, the build surface **30** may be considered to be oriented parallel to an X-Y plane of the apparatus **10**, and a direction perpendicular to the X-Y plane is denoted as a Z-axis direction (X, Y, and Z being three mutually perpendicular directions). As used herein, the X-axis refers to the machine direction along the length of the resin support **26**. As used herein, the Y-axis refers to the transverse direction across the width of the resin support **26** and generally perpendicular to the machine direction. As used herein, the Z-axis refers to the stage direction that can be defined as the direction of movement of the stage **18** relative to the window **16**.

(39) The build surface **30** may be configured to be “non-stick,” that is, resistant to adhesion of a cured resin R. The non-stick properties may be embodied by a combination of variables such as the chemistry of the resin support **26**, its surface finish, and/or applied coatings. For instance, a permanent or semi-permanent non-stick coating may be applied. One non-limiting example of a suitable coating is polytetrafluoroethylene (“PTFE”). In some examples, all or a portion of the build surface **30** may incorporate a controlled roughness or surface texture (e.g. protrusions, dimples, grooves, ridges, etc.) with nonstick properties. Additionally or alternatively, the resin support **26** may be made in whole or in part from an oxygen-permeable material.

(40) For reference purposes, an area or volume immediately surrounding the location of the resin support **26** and the window **16** or transparent portion defined by the support plate **14** may be defined as a “build zone,” labeled **32**.

(41) In some instances, a material depositor **34** may be positioned along the resin support **26**. The material depositor **34** may be any device or combination of devices that is operable to apply a layer of resin R on the resin support **26**. The material depositor **34** may optionally include a device or

combination of devices to define a height of the resin R on the resin support **26** and/or to level the resin R on the resin support **26**. Nonlimiting examples of suitable material deposition devices include chutes, rollers, hoppers, pumps, spray nozzles, spray bars, or printheads (e.g. inkjets). In some examples, a doctor blade may be used to control the thickness of resin R applied to the resin support **26** as the resin support **26** passes the material depositor **34**.

(42) In the illustrated example of FIG. **1B**, the resin support **26** may be in the form of a vat **36** that is configured to isolate debris that could contaminate the build from usable resin R. The vat **36** may include a floor **38** and a perimeter wall **40**. The perimeter wall **40** extends from the floor **38**. Inner surfaces of the floor **38** and the perimeter wall **40** define a receptacle **42** for receiving the resin R.

(43) A drive system **28** (FIG. **1A**) may be provided for moving the vat **36** relative to the stage **18** parallel to the X-direction between a build zone **32** and a position at least partially external to the build zone **32**. However, it will be appreciated that, in other embodiments, the resin support **26** may be stationary without departing from the scope of the present disclosure.

(44) In some instances, the resin support **26** may be positioned to accept a resin R from a material depositor **34** is operable to introduce a layer of resin R into the resin support **26**. The material depositor **34** may optionally include a device or combination of devices to define a height in the resin and/or to level the resin R. Nonlimiting examples of suitable material deposition devices include chutes, hoppers, pumps, spray nozzles, spray bars, or printheads (e.g. inkjets).

(45) Referring back to FIGS. **1A** and **1B**, the resin R includes any radiant-energy curable material, which is capable of adhering or binding together the filler (if used) in the cured state. As used herein, the term “radiant-energy curable” refers to any material which solidifies or partially solidifies in response to the application of radiant energy of a particular frequency and energy level. For example, the resin R may include a photopolymer resin containing photo-initiator compounds functioning to trigger a polymerization reaction, causing the resin R to change from a liquid (or powdered) state to a solid state. Alternatively, the resin R may include a material that contains a solvent that may be evaporated out by the application of radiant energy. The uncured resin R may be provided in solid (e.g. granular) or liquid form, including a paste or slurry.

(46) Furthermore, the resin R can have a relatively high viscosity resin that will not “slump” or run off during the build process. The composition of the resin R may be selected as desired to suit a particular application. Mixtures of different compositions may be used. The resin R may be selected to have the ability to out-gas or burn off during further processing, such as a sintering process.

(47) Additionally or alternatively, the resin R may be selected to be a viscosity reducible composition. These compositions reduce in viscosity when a shear stress is applied or when they are heated. For example, the resin R may be selected to be shear-thinning such that the resin R exhibits reduced viscosity as an amount of stress applied to the resin R increases. Additionally or alternatively, the resin R may be selected to reduce in the viscosity as the resin R is heated.

(48) The resin R may incorporate a filler. The filler may be pre-mixed with resin R, then loaded into the material depositor **34**. Alternatively, the filler may be mixed with the resin R on the apparatus **10**. The filler includes particles, which are conventionally defined as “a very small bit of matter.” The filler may include any material that is chemically and physically compatible with the selected resin R. The particles may be regular or irregular in shape, may be uniform or non-uniform in size and may have variable aspect ratios. For example, the particles may take the form of powder, of small spheres or granules, or may be shaped like small rods or fibers.

(49) The composition of the filler, including its chemistry and microstructure, may be selected as desired to suit a particular application. For example, the filler may be metallic, ceramic, polymeric, and/or organic. Other examples of potential fillers include diamond, silicon, and graphite. Mixtures of different compositions may be used. In some examples, the filler composition may be selected for its electrical or electromagnetic properties, e.g. it may specifically be an electrical insulator, a dielectric material, an electrical conductor, and/or magnetic.

(50) The filler may be “fusible,” meaning it is capable of consolidation into a mass upon application of sufficient energy. For example, fusibility is a characteristic of many available powders including but not limited to polymeric, ceramic, glass, and metallic. The proportion of filler to resin R may be selected to

suit a particular application. Generally, any amount of filler may be used so long as the combined material is capable of flowing and being leveled, and there is sufficient resin R to hold together the particles of the filler in the cured state.

(51) The stage **18** is a structure defining a planar surface **44**, which is capable of being oriented parallel to the build surface **30** or the X-Y plane. Various devices may be provided for moving the stage **18** relative to the window **16**. For example, as illustrated in FIGS. **1A** and **1B**, the movement may be provided through an actuator assembly **46** that may be coupled with a static support **48**. In some embodiments, the actuator assembly **46** may include a first actuator **50** between the stage **18** and the static support **48** that allows for movement of the stage **18** in a first, vertical direction (e.g., along the Z-axis direction). The actuator assembly **46** may additionally or alternatively include a second actuator **52** between the stage **18** and the first actuator **50** and/or the static support **48** that allows for movement in the X-axis direction and/or the Y-axis direction. The actuator assembly **46** may additionally or alternatively include a third actuator **54** between the stage **18** and the second actuator **52** and/or the stage **18** that allows for movement in the X-axis direction and/or the Y-axis direction. The actuator assembly **46** may include any device practicable of moving the stage **18** in any direction, such as ballscrew electric actuators, linear electric actuators, pneumatic cylinders, hydraulic cylinders, delta drives, belt systems, or any other practicable device. It will be appreciated that, in other examples, the resin support may additionally or alternatively translate in the Y-axis direction (or any other direction).

(52) The radiant energy device **20** may be configured as any device or combination of devices operable to generate and project radiant energy at the resin R in a suitable pattern and with a suitable energy level and other operating characteristics to cure the resin R during the build process. For example, as shown in FIGS. **1A** and **1B**, the radiant energy device **20** may include a projector **56**, which may generally refer to any device operable to generate a radiant energy predetermined patterned image of suitable energy level and other operating characteristics to cure the resin R. As used herein, the term “patterned image” refers to a projection of radiant energy comprising an array of one or more individual pixels. Non-limiting examples of patterned image devices include a DLP projector or another digital micro-mirror device, a two-dimensional array of LEDs, a two-dimensional array of lasers, and/or optically addressed light valves. In the illustrated example, the projector **56** includes a radiant energy source **58** such as a UV lamp, an image forming apparatus **60** operable to receive a source beam **62** from the radiant energy source **58** and generate a patterned image **64** to be projected onto the surface of the resin R, and optionally focusing optics **66**, such as one or more lenses.

(53) The image forming apparatus **60** may include one or more mirrors, prisms, and/or lenses and is provided with suitable actuators, and arranged so that the source beam **62** from the radiant energy source **58** can be transformed into a pixelated image **64** in an X-Y plane coincident with the surface of the resin R. In the illustrated example, the image forming apparatus **60** may be a digital micro-mirror device.

(54) The projector **56** may incorporate additional components, such as actuators, mirrors, etc. configured to selectively move the image forming apparatus **60** or other part of the projector **56** with the effect of rastering or shifting the location of the patterned image **64** on the build surface **30**. Stated another way, the patterned image **64** may be moved away from a nominal or starting location.

(55) In addition to other types of radiant energy devices **20**, the radiant energy device **20** may include a “scanned beam apparatus” used herein to refer generally to any device operable to generate a radiant energy beam of suitable energy level and other operating characteristics to cure the resin R and to scan the beam over the surface of the resin R in a desired pattern. For example, the scanned beam apparatus can include a radiant energy source **58** and a beam steering apparatus. The radiant energy source **58** may include any device operable to generate a beam of suitable power and other operating characteristics to cure the resin R. Non-limiting examples of suitable radiant energy sources **58** include lasers or electron beam guns.

(56) In some instances, the apparatus **10** may include a material retention assembly **68** that may be configured to retain the resin support **26** in a predefined position along the support plate **14**. In some instances, the material retention assembly **68** can include one or more pneumatic actuation zones **70** with each pneumatic actuation zone **70** configured to selectively interact with the resin support **26** by

producing a force on a surface of the resin support **26** opposite the resin R.

(57) The one or more pneumatic actuation zones **70** may apply a negative pressure on a first surface of the resin support **26** that is opposite to the resin R, or a second side of the resin support **26**, to produce a suction or vacuum on the resin support **26**. The negative pressure may retain the resin support **26** in a desired position along the support plate **14**. The one or more pneumatic actuation zones **70** may also apply a positive pressure on the first surface of the resin support **26** that is opposite to the resin R, or a second side of the resin support **26**, to produce a pushing force on the resin support **26**. The positive pressure may release the resin support **26** from a component of the apparatus **10**, such as the window **16**, the material retention assembly **68**, etc. As used herein, a “negative” pressure is any pressure that is less than an ambient pressure proximate to one or more pneumatic actuation zones **70** such that fluid may be drawn into the one or more pneumatic actuation zones **70**. Conversely, a “positive” pressure is any pressure that is greater than an ambient pressure proximate to one or more pneumatic actuation zones **70** such that fluid may be exhausted from the one or more pneumatic actuation zones **70**. Further, a “neutral” pressure is any pressure that is generally equal to an ambient pressure proximate to one or more pneumatic actuation zones **70**.

(58) In some examples, the pneumatic actuation zones **70** may be fluidly coupled with a pneumatic assembly **72** through various hoses and one or more ports. The pneumatic assembly **72** may include any device capable of providing a vacuum/suction and/or pushing a fluid, such as air or a process gas (e.g., nitrogen or argon), through the one or more pneumatic actuation zones **70**. For instance, the pneumatic assembly **72** may include a pressurized fluid source that includes a compressor and/or a blower. The pneumatic assembly **72** may additionally or alternatively include any assembly capable of altering a pressure, such as a venturi vacuum pump. In some embodiments, one or more valves and/or switches may be coupled with the pneumatic assembly **72** and the one or more pneumatic actuation zones **70**. The one or more valves and/or switches are configured to regulate a pressure to each of the one or more pneumatic actuation zones **70**.

(59) In some embodiments, the pneumatic actuation zone **70** includes one or more apertures **74** of any size and shape for interacting with the resin support **26**. For instance, the apertures **74** may be any number and combination of holes, slits, or other geometric shapes defined by any component of the additive manufacturing apparatus **10**, such as a portion of the support plate **14**. Additionally, or alternatively, the apertures **74** may be defined by a portion of the support plate **14** being formed from a porous material, or through any other assembly in which a fluid may be moved from a first side of the support plate **14** to a second side of the support plate **14** to interact with the resin support **26**.

(60) In some examples, the pneumatic actuation zone **70** may be defined by a plenum **76**. The plenum **76** may be of any size and may be similar or varied from the shape of any remaining plenum **76**. In some instances, a gasket may be positioned about a rim of the plenum **76**. Additionally or alternatively, the material retention assembly **68** may include one or more clamps that compressively maintain the resin support **26** along the support plate **14**.

(61) With further reference to FIGS. **1A** and **1B**, a viscosity modification assembly **78** may be integrated within the support plate **14** and/or otherwise operably coupled with the resin support **26**. The viscosity modification assembly **78** may be configured to apply a shearing stress to the resin R to alter (e.g., reduce) a viscosity of the resin R. Additionally or alternatively, the viscosity modification assembly **78** may be configured to heat the resin R to alter the viscosity of the resin R. It will be appreciated that in embodiments that heat the resin R to alter the viscosity of the resin R, the heat provided may be within a predefined range that is sufficient to alter the viscosity of the resin R without causing any cross-linking in the polymer.

(62) In some embodiments, the viscosity modification assembly **78** may be configured to mechanically vibrate a portion of the support plate **14** to create a shearing stress on the resin R. For example, the viscosity modification assembly **78** may include a movement device **80** (e.g., a transducer) that is operably coupled with the support plate **14**. The movement device **80** may be configured to vibrate at least a portion of the support plate **14** or any other module of the apparatus **10** that is then transferred to the resin R. Additionally and/or alternatively, the movement device **80** may be configured to convert electrical energy to ultrasonic mechanical pressure waves that are transferred to the resin R. For

the movement device **80** may be in the form of an ultrasonic vibrating device, such as one utilizing a piezoelectric transducer. In other embodiments, the viscosity modification assembly **78**, in addition to or in lieu of the transducer, may include, alone or in conjunction with one or the other, a fluid, an acoustic, a motor (e.g., offset cam), a reciprocating piston, or any other movement device **80**.

(63) The movement device **80** may be operably coupled with the computing system **84**. The computing system **84** may include a signal generator that supplies an electric impulse to the movement device **80**, the voltage of which can be varied at different frequencies and with different waveshapes. The signal may, for example, be a pure sinusoidal wave or may be modulated with one or more other frequencies. Alternatively, the signal may be a stepped or spiked pulse. In some embodiments, the signal generator transmits a signal of between 20-80 kHz. For example, the signal is at about 60 kHz. The signal generator may, for example, transmit a constant amplitude signal at a constant frequency, or alternate one or both of these parameters. A power level can be selected as a percentage of maximum power.

(64) In other embodiments, the viscosity modification assembly **78** may be configured to create a shearing stress on the resin R through other configurations without departing from the scope of the present disclosure. For example, the viscosity modification assembly **78** may be configured as a probe that may be adjacent and in physical contact with the resin support **26** and/or any other module that may relay the shearing stress to the resin R on the resin support **26**. Additionally or alternatively, the viscosity modification assembly **78** may be configured as an ultrasonic or vibration plate that may be operably coupled with the resin support **26** and/or any other module of the apparatus **10** that may provide the shearing stress to the resin R on the resin support **26**.

(65) With further reference to FIGS. **1A** and **1B**, in various embodiments, a gasket **82** may be positioned between the window **16** and the support plate **14** to isolate movement of each of the window **16** and the support plate **14** from one another. By isolating movement of the window **16** from the support plate **14**, degradation issues of the apparatus **10** caused through the operation of viscosity modification assembly **78** may be mitigated. In various examples, the gasket **82** may be formed from a motion attenuating material, such as any of a wide variety of resilient elastomers including, but not limited to, materials containing natural rubber and silicone.

(66) As provided herein, in some instances, the viscosity modification assembly **78** may additionally or alternatively be capable of producing heat to alter the viscosity of the resin R. For example, fast heating processes, such as dielectric or microwave heating, can be used to avoid exposing the resin R to a long heating cycle before the temperature of use is reached.

(67) The computing system **84** in FIGS. **1A** and **1B** is a generalized representation of the hardware and software that may be implemented to control the operation of the apparatus **10**, including some or all of the stage **18**, the drive system **28**, the radiant energy device **20**, the actuator assembly **46**, the material retention assembly **68**, the viscosity modification assembly **78**, a movement device **80**, actuators, and the various parts of the apparatus **10** described herein. The computing system **84** may be embodied, for example, by software running on one or more processors embodied in one or more devices such as a programmable logic controller (“PLC”) or a microcomputer. Such processors may be coupled to process sensors and operating components, for example, through wired or wireless connections. The same processor or processors may be used to retrieve and analyze sensor data, for statistical analysis, and for feedback control. Numerous aspects of the apparatus **10** may be subject to closed-loop control.

(68) Optionally, the components of the apparatus **10** may be surrounded by a housing **86**, which may be used to provide a shielding or inert gas (e.g., a “process gas”) atmosphere using gas ports **88**. Optionally, the pressure within the housing **86** could be maintained at a desired level greater than or less than atmospheric. Optionally, the housing **86** could be temperature and/or humidity controlled. Optionally, ventilation of the housing **86** could be controlled based on factors such as a time interval, temperature, humidity, and/or chemical species concentration. In some embodiments, the housing **86** can be maintained at a pressure that is different than an atmospheric pressure.

(69) Referring to FIGS. **2-4**, various perspective views are provided of the additive manufacturing apparatus **10** including an actuator assembly **46** in accordance with exemplary embodiments of the present disclosure. The exemplary apparatus **10** may include a base structure **90** and a static support **48** extending from the base structure **90**. In the illustrated embodiment, the actuator assembly **46** is

operably coupled with the static support **48** and is configured to change a position of the stage **18**. However, it will be appreciated that the actuator assembly **46** may be operably coupled with any other component without departing from the scope of the present disclosure.

(70) As illustrated, the actuator assembly **46** includes the first actuator **50** operably coupled with the static support **48**. A second actuator **52** may be operably coupled with and moveable along the first actuator **50**. Additionally, a third actuator **54** may be operably coupled with and moveable with the second actuator **52**. The stage **18** may be operably coupled with and moveable along the third actuator **54**. In various embodiments, when the stage **18** is moved along the Z-axis direction through the use of the first actuator **50**, the second actuator **52**, the third actuator **54**, and the stage **18** move along the first actuator **50**. When the stage **18** is moved in the X-axis direction through the use of the second actuator **52**, the third actuator **54**, and the stage **18** move along the second actuator **52**. When the stage **18** is moved in the Y-axis direction, the stage **18** moves along the third actuator **54**. While the first actuator **50** is illustrated translating in the Z-direction, the second actuator **52** is illustrated translating in the X-axis direction, and the third actuator **54** is illustrated translating in the Y-direction in FIG. 2, it will be appreciated that each actuator may be coupled to and moveable with any other actuator and/or the stage **18** without departing from the scope of the present disclosure. It will be appreciated that the terms “first actuator,” “second actuator,” and “third actuator” are used herein for clarity purposes. It will be understood that the additive manufacturing apparatus may include any one or more of the actuators without departing from the scope of the present disclosure.

(71) With further reference to FIGS. 2-4, the resin support **26** may be configured to retain first and second materials thereon. For example, the first material may be a first resin R.sub.1 and the second material may be a second resin R.sub.2. As illustrated, the first resin R.sub.1 may be positioned adjacent to a second resin R.sub.2 in the Y-axis direction. As such, in some instances, the first resin R.sub.1 may be positioned over a first segment of the window **16** as the resin support **26** is translated across the window **16**. Likewise, the second resin R.sub.2 may be positioned over a second segment of the window **16** that is offset from the first segment as the resin support **26** is translated across the window **16**.

(72) In various embodiments, such as the ones illustrated in FIGS. 2-4, the resin support **26** may be in the form of a first resin support **26A** that is configured to have the first resin R.sub.1 deposited thereon and a second resin support **26B** that is configured to have the second resin R.sub.2 deposited thereon. Each of the first resin support **26A** and the second resin support **26B** may be operably coupled with the feed module **22** (FIG. 1A) and the take-up module **24** (FIG. 1A). In some instances, each of the first resin support **26A** and the second resin support **26B** may be operably coupled with a common first roller **22A** within the feed module **22** and/or a common second roller **24A** within the take-up module **24**. In such instances, a translational movement length of the first resin support **26A** may be generally equal to a translational length of the second resin support **26B**. However, it will be appreciated that the first resin support **26A** may be operably coupled with a different roller within the feed module **22** and the take-up module **24** from the second resin support **26B** such that each resin support **26A**, **26B** may be translated in the X-direction independently of one another.

(73) With further reference to FIGS. 2-4, the material depositor **34** may include a first reservoir **92** configured to retain the first resin R.sub.1 and a second reservoir **94** configured to retain the second resin R.sub.2. As illustrated, the first reservoir **92** may be configured to retain the first resin R.sub.1 and the second reservoir **94** may be configured to retain the second resin R.sub.2.

(74) In some examples, the material depositor **34** can further include a first vessel **96** fluidly coupled with the first reservoir **92**. A first conduit **98** extends from the first vessel **96** to direct the first resin R.sub.1 from the first vessel **96** to the first reservoir **92**. Likewise, the material depositor **34** can also include a second vessel **100** fluidly coupled with the second reservoir **94**. A second conduit **102** extends from the second vessel **100** to direct the second resin R.sub.2 from the second vessel **100** to the second reservoir **94**.

(75) In various embodiments, the material depositor **34** can further include a first volume sensor **104** and/or a second volume sensor **106**. The first volume sensor **104** can be configured to provide signals to the computing system **84** related to a volume of the first resin R.sub.1 within the first reservoir **92**. The

computing system **84** is configured to receive the monitoring signals and process such signals using predetermined algorithms to generate control signals for controlling a first regulator **108**, which may allow or restrict flow of the first resin R.sub.1 from the first vessel **96** to the first reservoir **92**. Likewise, the second volume sensor **106** can be configured to provide signals to the computing system **84** related to a volume of the second resin R.sub.2 within the second reservoir **94**. The computing system **84** is configured to receive the monitoring signals and process such signals using predetermined algorithms to generate control signals for controlling a second regulator **110**, which may allow or restrict flow of the second resin R.sub.2 from the second vessel **100** to the first reservoir **92**. In this manner, closed-loop control of the volume of the first resin R.sub.1 and the volume of the second resin R.sub.2 can be achieved. The first volume sensor **104** and the second volume sensor **106** may each be embodied as one or more imaging sensors or any other vision-based device. The first volume sensor **104** and the second volume sensor **106** may additionally and/or alternatively be configured as any other practicable proximity sensor, such as, but not limited to, an ultrasonic sensor, a radar sensor, a LIDAR sensor, or the like.

(76) With further reference to FIGS. 2-4, a first thickness assembly **112** can be used to control the thickness of the resin R applied to the resin support **26** as the first resin R.sub.1 is deposited on the resin support **26**. A second thickness assembly **114** can be used to control the thickness of the second resin R.sub.2 applied to the resin support **26** as the second resin R.sub.2 is deposited on the resin support **26**. In the illustrated embodiment, the thickness of the first resin R.sub.1 may be defined through the usage of a doctor blade and the thickness of the second resin R.sub.2 may be defined through the usage of a doctor blade. In various embodiments, other material depositing apparatuses can be used separately or in combination with the first and second doctor blades, such as but not limited to, gravure rolls, metering rolls, weir-based cascades, direct die casting, and a combination thereof.

(77) In some embodiments, a common thickness assembly may additionally or alternatively be used with the first thickness assembly **112** and the second thickness assembly **114**. For instance, a common material depositing apparatus can be used for each of the first resin R.sub.1 and the second resin R.sub.2. In some instances, the common material depositing apparatus may be configured to act as a gross control for the thickness of an initially deposited layer of the first resin R.sub.1 and the second resin R.sub.2. In turn, the first thickness assembly **112** may further define the thickness of the first resin R.sub.1 and the second thickness assembly **114** may further define the thickness of the second resin R.sub.2.

(78) Still referring to FIGS. 2-4, a radiant energy device **20** may be positioned within the base structure **90**. As an option, the radiant energy device **20** may be coupled with an image movement device **116** through a bracket **118**. The movement device **80** may include actuators, mirrors, etc. that are configured to selectively move the radiant energy device **20**, or another part of the radiant energy device **20**, with the effect of rastering or shifting the location of a patterned image **64** relative to the window **16**. Stated another way, the patterned image **64** may be moved away from a nominal or starting location. This permits a single radiant energy device **20** to cover a larger build area, for example. This type of image projection may be referred to herein as a “tiled image”.

(79) Additionally or alternatively, the apparatus **10** may include a plurality of radiant energy devices **20** that are operably coupled with the build zone **32**. Each of the plurality of radiant energy devices **20** may or may not be configured to translate below the window **16** and/or the support plate **14**. Moreover, each of the plurality of radiant energy devices **20** may generate an image **64** that at least partially overlaps with an image **64** of an additional radiant energy device to form a stitched image on the resin R. In various embodiments, the images **64** from each of the plurality of radiant energy devices **20** may have some degree of overlap where that overlap is a single pixel, less than one pixel (for example, half a pixel), or more than one pixel. Further, in some embodiments, optics **66** may be optically coupled with the one or more radiant energy devices **20**. In such instances, at least one of the one or more radiant energy devices **20** and/or the optics **66** may translate along the Y-axis and/or otherwise move through the movement device **80** to produce patterned images **64** on various portions of the resin support **26**.

(80) Further, in some embodiments, the apparatus **10** may include one or more sensors **120** that are configured to detect information related to a position of the stage **18**, the resin support **26**, and/or the

radiant energy device **20**. For example, the one or more sensors **120** may verify the position of the stage **18** and/or the resin support **26** each time the stage **18** and/or the resin support **26** are moved by the actuator assembly **46**. Likewise, the one or more sensors **120** may verify the location of the radiant energy device **20** each time the radiant energy device **20** is translated by the movement device **80**. In various embodiments, the one or more sensors **120** may be any combination of devices that is configured to provide information indicative of a position of the stage **18** and/or the resin support **26** or a location of the radiant energy device **20**. For example, the one or more sensors **120** may include a gyroscope, an accelerometer, a proximity sensor, an image sensor, and/or any other practicable sensor.

(81) In operation, the radiant energy device **20** and/or the movement assembly may produce heat. Accordingly, one or more vents and/or fans **122** may be positioned within the base structure **90** to remove heat from the base structure **90**. The one or more fans **122** may be configured as any fluid movement device **80** that is capable of drawing the heated air from the base structure **90** to an area proximate to the base structure **90**.

(82) Referring now to FIG. 5, a method **200** for operating an additive manufacturing apparatus **10** is provided in accordance with various aspects of the present disclosure. The various steps of method **200** are schematically illustrated in FIGS. 5-24. The method **200** can be used to operate the additive manufacturing apparatus **10** or any other suitable additive manufacturing apparatus **10**. It should be appreciated that the example method **200** is discussed herein only to describe example aspects of the present subject matter and is not intended to be limiting. Any of the steps within FIG. 5 may be omitted without departing from the scope of the present disclosure.

(83) Referring now to FIGS. 5 and 6, the method **200** can include, at step **202**, depositing a layer of a first uncured resin and a second uncured resin onto the resin support **26**. In various embodiments, the first resin R.sub.1 may be laterally offset from the second resin R.sub.2 on the resin support **26** in the Y-direction. Further, a gap **124** may be defined between the laterally offset first and second resins R.sub.1, R.sub.2.

(84) In addition, the resin support **26** may be in the form of a first resin support **26A** that is configured to have the first resin R.sub.1 deposited thereon and a second resin support **26B** that is configured to have the second resin R.sub.2 deposited thereon. Each of the first resin support **26A** and the second resin support **26B** may be operably coupled with the feed module **22** and the take-up module **24**. Conversely, the first resin R.sub.1 and the second resin R.sub.2 may be deposited on a common resin support **26**.

(85) While the first resin R.sub.1 and/or the second resin R.sub.2 is deposited onto the resin support **26** and/or after the deposition of the first resin R.sub.1 and/or the second resin R.sub.2 onto the resin support **26**, the resin support **26** may be translated in an X-axis direction into a build zone **32**.

(86) As shown in FIGS. 5 and 7, at step **204**, the method **200** can include placing the stage **18** in a first curing position by moving the stage **18** such that a working surface of the stage **18** and/or the component **12** retained by the stage **18** contacts the first resin R.sub.1. As the stage **18** is moved to the first curing position, a working surface of the stage **18** and/or the component **12** retained by the stage **18** contacts the first resin R.sub.1.

(87) As shown in FIGS. 5 and 8, at step **206**, the method **200** can include curing a portion of the first resin R.sub.1 while the stage **18** is in the first curing position relative to a window **16** by applying radiant energy from a radiant energy device **20** through the window **16** and the resin support **26**. As provided herein, the radiant energy may be in the form of a first patterned image **64** that is transmitted through at least a portion of the window **16**. The portion of the first resin R.sub.1 that is cured forms a layer of the component **12** that is retained by the stage **18**. In the exemplary embodiment of FIG. 8, the patterned image **64** is generally equal to the width of the first resin R.sub.1 in the Y-axis direction. However, the first portion may be of any width (varied or constant) in the Y-axis direction.

(88) As shown in FIGS. 5 and 9, at step **208**, the method **200** can include separating the component **12** from the resin support **26** by altering a position of the stage **18** through the actuator assembly **46**. It will be appreciated that the stage **18** may move in the X-axis direction, the Y-axis direction, and/or the Z-axis direction.

(89) As shown in FIGS. 5 and 10, at step **210**, the method **200** can include placing the stage **18** in a

second curing position by moving the stage **18** such that the working surface of the component **12** retained by the stage **18** contacts the second resin R.sub.2. As the stage **18** is moved to the second curing position, the working surface of the component **12** retained by the stage **18** contacts the second resin R.sub.2.

(90) As shown in FIGS. **5** and **11**, at step **212**, the method **200** can include curing a portion of the second resin R.sub.2 while the stage **18** is in the second curing position relative to a window **16** by applying radiant energy from a radiant energy device **20** through the window **16** and the resin support **26**. As provided herein, the radiant energy may be in the form of a patterned image **64** that is transmitted through at least a portion of the window **16**. The portion of the second resin R.sub.2 that is cured forms a layer of the component **12** that is retained by the stage **18**. In the exemplary embodiment of FIG. **11**, the patterned image **64** is generally equal to the width of the second resin R.sub.2 in the Y-axis direction. However, the second portion may be of any width (varied or constant) in the Y-axis direction.

(91) As shown in FIGS. **5** and **12**, at step **214**, the method **200** can include separating the component **12** from the resin support **26** by altering a position of the stage **18** through the actuator assembly **46**. It will be appreciated that the stage **18** may move in the X-axis direction, the Y-axis direction, and/or the Z-axis direction.

(92) As shown in FIGS. **5** and **13**, at step **216**, after the component **12** is separated from the resin support **26**, the method **200** can include translating the resin support **26** such that a fresh first resin R.sub.1 and a second resin R.sub.2 are positioned between the window **16** and the stage **18**. It will be appreciated, however, that in various embodiments, after a portion of either resin is cured, the resin support **26** supporting that respective resin may be translated. Additionally or alternatively, the resin support **26** may be translated whenever the next layer to be formed requires fresh resin. For example, if the first resin R.sub.1 forms a first layer and a second resin R.sub.2 forms a second layer, the resin support **26** may be translated after the second layer. If the first resin R.sub.1 forms a first layer and a second layer, the resin support **26** may be translated between the first and second layers even though the second resin R.sub.2 remains unused between the first and second layers.

(93) While the resin support **26** is translated, the stage **18** and the component **12** attached to the stage **18** may move based on the next layer to be formed. For example, in the illustrated embodiment, the third layer will include the first resin R.sub.1 therein. As such, the stage **18** may be moved in the Y-axis direction to a position above the first resin R.sub.1.

(94) As shown in FIGS. **5** and **14**, at step **218**, the method **200** can include placing the stage **18** in a third curing position by moving the stage **18** such that the working surface of the component **12** retained by the stage **18** contacts the first resin R.sub.1. As the stage **18** is moved to the third curing position, the working surface of the component **12** retained by the stage **18** contacts the first resin R.sub.1.

(95) As shown in FIGS. **5** and **15**, at step **220**, the method **200** can include curing a portion of the first resin R.sub.1 while the stage **18** is in the third curing position relative to a window **16** by applying radiant energy from a radiant energy device **20** through the window **16** and the resin support **26**. As provided herein, the radiant energy may be in the form of a patterned image **64** that is transmitted through at least a portion of the window **16**. The portion of the first resin R.sub.1 that is cured forms a layer of the component **12** that is retained by the stage **18**. In the exemplary embodiment of FIG. **15**, the patterned image **64** is less than the width of the first resin R.sub.1 in the Y-axis direction. However, the cured portion may be of any width (varied or constant) in the Y-axis direction.

(96) As shown in FIGS. **5**, **16**, and **17**, at step **222**, the method **200** can include separating the component **12** from the resin support **26** by altering a position of the stage **18** through the actuator assembly **46**. It will be appreciated that the stage **18** may move in the X-axis direction, the Y-axis direction, and/or the Z-axis direction.

(97) As shown in FIGS. **5** and **18**, at step **224**, the method **200** can include placing the stage **18** in a fourth curing position by moving the stage **18** such that the working surface of the component **12** retained by the stage **18** contacts the second resin R.sub.2. As the stage **18** is moved to the fourth curing position, the working surface of the component **12** retained by the stage **18** contacts the second resin R.sub.2.

(98) As shown in FIGS. 5 and 19, at step 226, the method 200 can include curing a portion of the second resin R.sub.2 while the stage 18 is in the fourth curing position relative to a window 16 by applying radiant energy from a radiant energy device 20 through the window 16 and the resin support 26. As provided herein, the radiant energy may be in the form of a patterned image 64 that is transmitted through at least a portion of the window 16. The portion of the second resin R.sub.2 that is cured forms a layer of the component 12 that is retained by the stage 18. In the exemplary embodiment of FIG. 19, the patterned image 64 is less than the width of the second resin R.sub.2 in the Y-axis direction. However, the cured portion may be of any width (varied or constant) in the Y-axis direction.

(99) As shown in FIGS. 5 and 20, at step 230, the method 200 can include separating the component 12 from the resin support 26 by altering a position of the stage 18 through the actuator assembly 46. It will be appreciated that the stage 18 may move in the X-axis direction, the Y-axis direction, and/or the Z-axis direction.

(100) As shown in FIGS. 5 and 21, at step 232, after the component 12 is separated from the resin support 26, the method 200 can include translating the resin support 26 such that a fresh first resin R.sub.1 and a second resin R.sub.2 are positioned within the build zone 32.

(101) As shown in FIGS. 5 and 22, at step 234, the method 200 can include placing the stage 18 in a fifth curing position by moving the stage 18 in the X-axis direction, Y-axis direction, and/or a Z-axis direction such that the working surface of the component 12 is retained by the stage 18 contacts both the first resin R.sub.1 and the second resin R.sub.2. As the stage 18 is moved to the fifth curing position, the working surface of the component 12 retained by the stage 18 contacts the first resin R.sub.1 and the second resin R.sub.2.

(102) As shown in FIGS. 5 and 23, at step 236, the method 200 can include curing a portion of the first resin R.sub.1 and the second resin R.sub.2 while the stage 18 is in the fifth curing position relative to a window 16 by applying radiant energy from a radiant energy device 20 through the window 16 and the resin support 26. As provided herein, the radiant energy may be in the form of a patterned image 64 that is transmitted through at least a portion of the window 16.

(103) As shown in FIGS. 5 and 24, at step 238, the method 200 can include separating the component 12 from the resin support 26 by altering a position of the stage 18 through the actuator assembly 46. It will be appreciated that the stage 18 and the resin support 26 may move to any number (one or more) curing positions to form each layer of the component 12 based on the component design.

(104) Referring now to FIGS. 25-34, various views of the additive manufacturing apparatus 10 are illustrated according to various aspects of the present disclosure. While only some components of the additive manufacturing apparatus are illustrated in FIGS. 25-34 for clarity purposes, it will be appreciated that the additive manufacturing apparatus may include any feature described in the present disclosure. In addition, it will be appreciated that the additive manufacturing apparatus may perform the method 200 described in FIGS. 5-24 and/or any other method, such as method 400 set forth in FIG. 36.

(105) In the embodiments illustrated in FIGS. 25-34, the additive manufacturing apparatus may include a frame 126 and an actuator assembly 46 configured to allow for movement of various components of the additive manufacturing apparatus relative to one another. In some instances, the frame 126 may include a frame structure 128 and may be operably coupled with and/or operably support various components of the additive manufacturing apparatus. As illustrated, the frame 126 may support a build plate 130, a support plate 14, and/or a base plate 132.

(106) The build plate 130 may support a first actuator 50. In some embodiments, the first actuator 50 can allow for movement of the stage 18 in a first, vertical direction (e.g., along the Z-axis direction). The first actuator 50 may include any device practicable of moving the stage 18 in the Z-axis direction, such as ballscrew electric actuators, linear electric actuators, pneumatic cylinders, hydraulic cylinders, delta drives, belt systems, or any other practicable device.

(107) A print head 134 may be operably coupled with the first actuator 50. In such instances, the print head 134 may be configured to translate or otherwise through actuation of the first actuator 50. The print head 134 may further be operably coupled with the stage 18. In some instances, the print head 134 may include a clamp 136 and/or any other feature for retaining the stage 18 on the print head 134.

(108) The frame 126 may further be operably coupled with the support plate 14. In various

embodiments, a window **16** can be operably supported by the support plate **14**. Further, the window **16** and the support plate **14** can be integrally formed such that one or more windows **16** are integrated within the support plate **14**.

(109) With further reference to FIGS. **25-34**, the base plate **132** may be fixed to a bottom portion of the frame structure **128**. A first slide assembly **138** may be operably coupled with the base plate **132**. The first slide assembly **138** may be configured to guide movement of the base plate **132** relative to a mounting plate **140**, which may be retained within a generally static position within the additive manufacturing apparatus. It will be appreciated, however, that the mounting plate **140** may be moveable.

(110) In the embodiments illustrated in FIGS. **25-34**, one or more guides **142** may be operably coupled with the base plate **132**. The guides **142** may move along a track **144** operably coupled with the mounting plate **140**. As will be described in greater detail below, the first slide assembly **138** may allow for the frame **126**, and, consequently, the stage **18** to translate along the track **144** in the Y-axis direction such that the stage **18** may be lowered onto a first resin R.sub.1 and/or a second resin R.sub.2. As provided herein, the first resin R.sub.1 may be separated from the second resin R.sub.2 by a gap **124**. As such, the stage **18** may be moved to the various resins to form various portions of the component **12**. It will be appreciated that in other embodiments, the track **144** may be operably coupled with the base plate **132** and the guides **142** may be operably coupled with the mounting plate **140** without departing from the scope of the present disclosure. In addition, it will be appreciated that the first slide assembly **138** may include any components that allow for movement of the base plate **132** relative to the mounting plate **140** without departing from the teachings provided herein.

(111) In various embodiments, a second actuator **52** may be operably coupled with the base plate **132** and the mounting plate **140**. The second actuator **52** may be configured to move the base plate **132** relative to the mounting plate **140** in the Y-axis direction. The second actuator **52** may include any practicable device, such as ballscrew electric actuators, linear electric actuators, pneumatic cylinders, hydraulic cylinders, delta drives, belt systems, or any other practicable device.

(112) Referring still to FIGS. **25-34**, in various embodiments, the radiant energy device **20** may be operably coupled with the frame **126** and positioned on an opposing side of the window from the stage **18** in the Z-direction. In the illustrated embodiments, the radiant energy device **20** may include a pair of projectors **56**. However, it will be appreciated that the radiant energy device **20** may be configured as any device or combination of devices operable to generate and project radiant energy at the resin R in a suitable pattern and with a suitable energy level and other operating characteristics to cure the resin R during a build process.

(113) In some embodiments, the radiant energy device **20** may be operably coupled with a carrier plate **146**. In some instances, the radiant energy device **20** may include one or more carrier rails **148** that are coupled with the carrier plate **146** for retaining the radiant energy device **20** relative to the carrier plate **146**.

(114) In several embodiments, the radiant energy device **20** may be movably coupled with the frame **126**. For example, a second slide assembly **150** may be positioned between the carrier plate **146** and the base plate **132**. The second slide assembly **150** may be configured to guide movement of the carrier plate **146** relative to the base plate **132**. As such, in some examples, the stage **18** and the radiant energy device **20** may move in conjunction relative to the mounting panel with one another through the use of the first slide assembly **138** and the radiant energy device **20** may move relative to the stage **18** through use of the second slide assembly **150**.

(115) In the embodiments illustrated in FIGS. **25-34**, one or more guides **152** may be operably coupled with the carrier plate **146**. The guides **152** may move along a rail **154** operably coupled with the base plate **132**. As will be described in greater detail below, the second slide assembly **150** may allow for the radiant energy device **20** to translate along the second slide assembly **150** in the Y-axis direction such that the radiant energy device **20** may be aligned with various portions of the resin support **26**. It will be appreciated that in other embodiments, the rail **154** may be operably coupled with the carrier plate **146** and the guides **152** may be operably coupled with the base plate **132** without departing from the scope of the present disclosure. In addition, it will be appreciated that the second slide assembly **150** may be

operably coupled with any other component, such as the frame **126** or the mounting plate **140** without departing from the scope of the present disclosure.

(116) In various embodiments, a third actuator **54** may be operably coupled with the carrier plate **146** and the base plate **132**. The third actuator **54** may be configured to move the carrier plate **146** relative to the base plate **132** in the Y-axis direction. The third actuator **54** may include any practicable device, such as ballscrew electric actuators, linear electric actuators, pneumatic cylinders, hydraulic cylinders, delta drives, belt systems, or any other practicable device.

(117) As illustrated in FIG. **25**, a cover **156** may be operably coupled with the frame **126** and configured to shield an area between the lenses of the radiant energy device **20** and the support plate **14**. In various embodiments, the shield may be configured to prevent light within various spectrums and/or all light from passing therethrough.

(118) Referring further to FIGS. **28** and **29**, the print head **134** may include a print head mounting plate **158**, one or more print head arms **160** extending from the print head mounting plate **158**, and an attachment assembly **162**. The print head **134** may be configured to selectively retain the stage **18**.

(119) In some embodiments, such as the one illustrated in FIG. **29**, the stage **18** is a structure defining a planar surface **44**, which is capable of being oriented parallel to a build surface **30** or the X-Y plane. The stage **18** may further include one or more stage rails **164** that are configured to interact with the print head **134**. One or more datums **166** may also be positioned on and/or integrally formed with the stage **18**. In some instances, the stage **18** may further include one or more handles **168**.

(120) Referring back to FIGS. **28** and **29**, the clamp **136** of the attachment assembly **162** may selectively retain the stage **18** relative to the print head **134**. In addition, the attachment assembly **162** may further include an anchor plate **170** and a clamping plate **172**. In operation, the anchor plate **170** may maintain a common position when the clamp **136** is in both the unlocked and locked positions. The clamping plate **172** may be operably coupled with a clamping arm **174**. The clamping plate **172** and the clamping arm **174** may move in conjunction with one another when the clamp **136** is moved from the unlocked position to the locked position and vice versa.

(121) The clamping arm **174** may include one or more protrusions **176** that are configured to be positioned within the one or more rails **154** of the stage **18**. In some instances, the one or more protrusions **176** may define notches **178** for assisting in locating the stage **18** relative to the protrusions **176**. With the one or more protrusions **176** within the one or more rails **154** of the stage **18**, the one or more datums **166** of the stage **18** can align with locators **180** within the attachment assembly **162**. In various embodiments, as the clamp **136** is moved to the locked position, a distance between a bottom portion of the locators **180** and a top portion of the clamping arm **174** is increased in the Z-axis direction. The increased distance creates an expansion force on the rails **154** of the stage **18** and the one or more datums **166** thereby retaining the stage **18** in position relative to the print head **134**.

(122) Referring further to FIGS. **30** and **31**, as provided herein, the mounting plate **140** may have a pair of tracks **144** operably coupled thereto. A plurality of guides **142** are coupled with the base plate **132** and may be configured to move along the first slide assembly **138**. As illustrated, the second actuator **52** may be operably coupled with the base plate **132** and the mounting plate **140** to move the base plate **132** relative to the mounting plate **140**. In some instances, the second actuator **52** may be configured to move the base plate **132** between a first position and a second position.

(123) In the first position, the stage **18** may be lowered by the first actuator **50** (FIG. **25**) to a position over at least a portion of the first resin R.sub.1. When in the first position, the guides **142** may contact a first pair of stops **182**. In the second position, the stage **18** may be lowered by the first actuator **50** (FIG. **25**) to a position over at least a portion of the second resin R.sub.2. When in the second position, the guides **142** may contact a second pair of stops **184**. It will be appreciated that the first slide assembly **138** may include a single first stop and/or a single second stop without departing from the scope of the present disclosure.

(124) In various embodiments, each of the first pair of stops **182** and the second pair of stops **184** may be adjustable. For example, the first pair of stops **182**, and the second pair of stops **184** can be in the form of fine threaded (e.g., **100** threads per inch (TPI)) adjustable end stops that allow precise positioning of each respective pair of end stops **182**, **184**. In some embodiments, the one or more guides

142 may include standoffs **186** that are configured to engage with end stops **182, 184**. The standoffs **186** can be removable and replaceable thereby allowing for easy replacement once the standoffs **186** in use are worn from use and/or for any other reason.

(125) With further reference to FIG. **31**, as provided herein, a second slide assembly **150** may include a pair of rails **154** operably coupled to the base plate **132**. The second slide assembly **150** may also include one or more guides **152** coupled with the carrier plate **146**. The guides **152** may be configured to move along the pair of rails **154**. As illustrated, the third actuator **54** may be operably coupled with the base plate **132** and the carrier plate **146** to move the carrier plate **146** relative to the base plate **132**. In some instances, the third actuator **54** may be configured to move the carrier plate **146** between a first position and a second position.

(126) In the first position, the radiant energy device **20** may be positioned in a first location relative to the window **16** and/or the base plate **132**. When in the first position, the guides **152** may contact a first stop **188**. In the second position, the radiant energy device **20** may be positioned in a second location relative to the window **16** and/or the base plate **132** that is offset from the first location in the Y-axis direction. When in the second position, the guides **152** may contact a second stop **190**.

(127) In the illustrated embodiment of FIG. **31**, the first stop **188** may be operably coupled to the base plate **132**. The second stop **190** may be operably coupled with a flange **192** of the base plate **132**. In various embodiments, each of the first end stop and the second end stop may be adjustable. For example, the first end stop, and the second end stop can be in the form of a fine threaded (e.g., **100 TPI**) adjustable end stops that allow for precise positioning of each of the first end stop and the second end stop. In some embodiments, a first standoff **186** may be supported by the carrier plate **146** and configured to align with the first end stop. A second standoff **186** may be operably coupled with an opposing end of the carrier plate **146** and/or the guides **152** of the second slide assembly **150**. Each of the first standoff **186** and the second standoff **186** can be removable and replaceable thereby allowing for easy replacement once the standoffs **186** in use are worn from use and/or for any other reason.

(128) Referring now to FIGS. **32-34**, various perspective views of the additive manufacturing apparatus having the actuator assembly **46** are provided in accordance with various aspects of the present disclosure. As provided herein, the additive manufacturing apparatus may be configured to deposit a first resin R.sub.1 and a second resin R.sub.2 onto a resin support **26**. The first resin R.sub.1 may be laterally offset from the second resin R.sub.2 on the resin support **26** in the Y-direction. In addition, a gap **124** may be defined between the laterally offset first and second resins R.sub.1, R.sub.2.

(129) The actuator assembly **46** may include a first actuator **50** configured to move the stage **18** in a plurality of locations along the Z-axis direction. The actuator assembly **46** may also include a second actuator **52** that may be associated with the first slide assembly **138**. The first slide assembly **138** may be configured to move the stage **18** relative to the mounting plate **140**. In addition, the actuator assembly **46** may include a third actuator **54** that is may be associated with the second slide assembly **150**. The second slide assembly **150** may be configured to move the radiant energy device **20** relative to the frame **126** and/or the window **16**.

(130) During operation, the resin support **26** may be maintained in a generally consistent position. Conversely, the stage **18** and/or the radiant energy device **20** may be movable relative to the resin support **26** such that the first resin R.sub.1 (or various portions of the first resin R.sub.1) and/or the second resin R.sub.2 (or various portions of the second resin R.sub.2) may be selectively cured.

(131) For example, as illustrated in FIG. **32**, the first slide assembly **138** may be placed in a first position relative to the mounting plate **140**. In the first position, the guides **142** of the first slide assembly **138** may contact and/or be proximate to the first pair of stops **182**. In the first position, the stage **18** may be generally aligned with the first resin R.sub.1 in the Y-axis direction.

(132) As also shown in FIG. **32**, the second slide assembly **150** may be in the first position. In the first position, the guides **152** of the second slide assembly **150** may contact and/or be proximate to the first stop **188** of the second slide assembly **150**. In the first position, the radiant energy device **20** may be in a first location relative to the window **16**. As such, the radiant energy device **20** may be directed towards a first portion of the first resin R.sub.1.

(133) As shown in FIG. **33**, while the first slide assembly **138** is in the first position, the second slide

assembly **150** may alternatively be in the second position. In the second position, the guides **152** of the second slide assembly **150** may contact and/or be proximate to the second stop **190** of the second slide assembly **150**. In the second position, the radiant energy device **20** may be in a second location relative to the window **16**. As such, the radiant energy device **20** may be directed towards a second portion of the first resin R.sub.1.

(134) As illustrated in FIG. **34**, the first slide assembly **138** may also be placed in a second position relative to the mounting plate **140**. In the second position, the guides **142** of the first slide assembly **138** may contact and/or be proximate to the second pair of stops **184**. In the second position, the stage **18** may be generally aligned with the second resin R.sub.2 in the Y-axis direction.

(135) As also shown in FIG. **34**, the second slide assembly **150** may be in the first position. In the first position, the guides **152** of the second slide assembly **150** may contact and/or be proximate to the first stop **188** of the second slide assembly **150**. In the first position, the radiant energy device **20** may be in a third location relative to the window **16**. As such, the radiant energy device **20** may be directed towards a first portion of the second resin R.sub.2.

(136) As shown in FIG. **35**, while the first slide assembly **138** is in the second position, the second slide assembly **150** may alternatively be in the second position. In the second position, the guides **152** of the second slide assembly **150** may contact and/or be proximate to the second stop **190** of the second slide assembly **150**. In the second position, the radiant energy device **20** may be in a fourth location relative to the window **16**. As such, the radiant energy device **20** may be directed towards a second portion of the second resin R.sub.2.

(137) During operation, each of the slide assemblies may be manipulated in any order to form a component **12** formed from one or both of the first and second resins R.sub.1, R.sub.2. In some instances, the radiant energy device **20** may be configured as any device or combination of devices operable to generate and project radiant energy at the resin R in a suitable pattern and with a suitable energy level and other operating characteristics to cure the first resin R.sub.1 and/or the second resin R.sub.2 during the build process. In addition, the radiant energy device **20** may or may not be configured to translate below the window **16** and/or the support plate **14**. For example, as shown in FIGS. **31-34**, the radiant energy device **20** may include a first projector **56** and a second projector **56**. Each of the first projector **56** and the second projector **56** may generate an image **64** that at least partially overlaps with an image **64** of an additional radiant energy device to form a stitched image **64** on the first resin R.sub.1 and/or the second resin R.sub.2. In various embodiments, the images **64** from each of the plurality of radiant energy devices **20** may have some degree of overlap where that overlap is a single pixel, less than one pixel (for example, half a pixel), or more than one pixel. Additionally or alternatively, the radiant energy device **20** may be capable of performing a scanning process in which the consecutive patterned images **64** are emitted from the radiant energy device **20** as the radiant energy device **20** is translated along the movement device **80**. As such, in some embodiments, the image **64** from the first projector **56** and the image **64** from the second projector **56** may be statically, or mechanically, stitched together. Additionally or alternatively, multiple images **64** from the first projector **56** may be stitched with one another as the third actuator **54** alters a position of the first projector **56**. Likewise, multiple images **64** from the second projector **56** may be stitched with one another as the third actuator **54** alters a position of the second projector **56**.

(138) Further, in some embodiments, the apparatus **10** may include one or more sensors **194** that are configured to detect information related to a position of the printhead, the frame **126**, and/or the radiant energy device **20**. For example, the one or more sensors **194** may verify the position of the stage **18** each time the stage **18** is moved. Additionally or alternatively, the one or more sensors **194** may verify the location of the frame **126** each time the frame **126** is translated. Additionally or alternatively, the one or more sensors **194** may verify the location of the radiant energy device **20** each time the radiant energy device **20** is translated by the movement device **80**. In various embodiments, the one or more sensors **194** may be any combination of devices that is configured to provide information indicative of a position of the printhead, the frame **126**, and/or the radiant energy device **20**. For example, the one or more sensors **194** may include a gyroscope, an accelerometer, a proximity sensor, an image sensor, and/or any other practicable sensor.

(139) Referring now to FIG. 36, a perspective view of the additive manufacturing apparatus is provided in accordance with various aspects of the present disclosure. As provided herein, the additive manufacturing apparatus may be configured to deposit a first resin R.sub.1 and a second resin R.sub.2 onto a resin support 26. The first resin R.sub.1 may be laterally offset from the second resin R.sub.2 on the resin support 26 in the Y-direction. In addition, a gap may be defined between the laterally offset first and second resins R.sub.1, R.sub.2.

(140) As illustrated, in various embodiments, the frame 126 may be movable relative to the support plate 14 and the window 16. As such, in some instances, there may be little to no relative motion between the resin support 26 and the support plate 14 and the window 16 in the Y-axis direction.

(141) In some embodiments, an opening 196 may be defined between the support plate 14 and the frame 126 to accommodate for movement of the frame 126 relative to the support plate 14. The opening 196 may have a length in the Y-axis direction that is at least equal to the length of travel of the build plate 130 when the frame 126 is moved from the first position to the second position.

(142) Referring now to FIG. 37, a method 400 for operating an additive manufacturing apparatus 10 is provided. The method 400 can be used to operate the additive manufacturing apparatus 10 or any other suitable additive manufacturing apparatus 10. It should be appreciated that the example method 400 is discussed herein only to describe example aspects of the present subject matter and is not intended to be limiting. Any of the steps within FIG. 36 may be omitted without departing from the scope of the present disclosure.

(143) At 402, the method 400 can include depositing a layer of a first uncured resin and a second uncured resin onto the resin support. In various embodiments, the first resin may be laterally offset from the second resin R.sub.2 on the resin support in the Y-direction. Further, a gap may be defined between the laterally offset first and second resins R.sub.1, R.sub.2.

(144) In addition, the resin support may be in the form of a first resin support that is configured to have the first resin deposited thereon and a second resin support that is configured to have the second resin deposited thereon. Each of the first resin support and the second resin support may be operably coupled with the feed module and the take-up module. Conversely, the first resin and the second resin may be deposited on a common resin support.

(145) At 404, the method 400 can include translating the resin support in an X-axis direction into a build zone while the first resin and/or the second resin is deposited onto the resin support and/or after the deposition of first resin and/or second resin onto the resin support. As provided herein, the resin support may be translated by a drive assembly.

(146) At 406, the method 400 can include placing the stage in a first curing position by moving the stage such that a working surface of the stage and/or the component retained by the stage contacts the first resin. As provided herein, the additive manufacturing apparatus includes a frame. The frame may include a frame structure, a build plate, a support plate, and/or a base plate.

(147) The build plate may support a first actuator. In some embodiments, the first actuator can allow for movement of the stage in a first, vertical direction (e.g., along the Z-axis direction). A print head may be operably coupled with the first actuator. In such instances, the print head may be configured to translate or otherwise be moved by the actuator in the Z-axis direction. The print head may further be operably coupled with the stage.

(148) The base plate may be fixed to a bottom portion of the frame structure. A first slide assembly may be operably coupled with the base plate. The first slide assembly may guide the movement of the base plate relative to a mounting plate in a Y-axis direction. The mounting plate may be retained within a generally static position within the additive manufacturing apparatus. In various embodiments, a second actuator may be operably coupled with the base plate and the mounting plate. The second actuator may be configured to move the base plate relative to the mounting plate in the Y-axis direction.

(149) At 408, the method 400 can include positioning a radiant energy device in a first projection position. In various embodiments, the radiant energy device may be operably coupled with the frame and positioned on an opposing side of the window from the stage in the Z-direction. In several embodiments, the radiant energy device may be movably coupled with the frame. For example, a second slide assembly may be positioned between the carrier plate and the base plate. The second slide

assembly may guide the movement of the carrier plate relative to the base plate. As such, in some examples, the stage and the radiant energy device may move in conjunction relative to the mounting panel with one another through the use of the first slide assembly and the radiant energy device may move relative to the stage through the use of the second slide assembly.

(150) At **410**, the method **400** can include curing a portion of the first resin while the stage is in the first curing position and the radiant energy device in the first projection position by applying radiant energy from a radiant energy device through a window of the support plate and the resin support. As provided herein, the radiant energy may be in the form of a patterned image that is transmitted through at least a portion of the window.

(151) At **412**, the method **400** can include positioning the radiant energy device in a second projection position. The second projection position is offset from the first curing position in the Y-axis direction. In various embodiments, the radiant energy device is moved from the first projection position to the second projection position by translating along a second slide assembly. A third actuator may be operably coupled with the base plate of the frame and a carrier plate operably coupled with the radiant energy device to move the radiant energy device relative to the stage and/or the resin support.

(152) At **414**, the method **400** can include curing a portion of the first resin while the stage is in the first curing position and the radiant energy device in the second projection position by applying radiant energy from a radiant energy device through a window of the support plate and the resin support. As provided herein, the radiant energy may be in the form of a patterned image that is transmitted through at least a portion of the window.

(153) At **416**, the method **400** can include placing the stage in a second curing position by moving the stage such that a working surface of the stage and/or the component retained by the stage contacts the first resin and/or the second resin. The second curing position is offset from the first curing position in the Y-axis direction.

(154) At **418**, the method **400** can include positioning a radiant energy device in a first projection position. In turn, at **420**, the method **400** can include curing a portion of the second resin while the stage is in the second curing position and the radiant energy device in the first projection position by applying radiant energy from a radiant energy device through a window of the support plate and the resin support. As provided herein, the radiant energy may be in the form of a patterned image that is transmitted through at least a portion of the window.

(155) At **422**, the method **400** can include positioning the radiant energy device in a second projection position. The second projection position is offset from the first curing position in the Y-axis direction. In various embodiments, the radiant energy device is moved from the first projection position to the second projection position by translating along the second slide assembly.

(156) At **424**, the method **400** can include curing a portion of the second resin while the stage is in the second curing position and the radiant energy device in the second projection position by applying radiant energy from a radiant energy device through a window of the support plate and the resin support.

(157) FIG. **38** depicts certain components of a computing system **84** according to example embodiments of the present disclosure. The computing system **84** can include one or more computing device(s) **84A** which may be used to implement the methods **200**, **400** described herein. The computing device(s) **84A** can include one or more processor(s) **84B** and one or more memory device(s) **84C**. The one or more processor(s) **84B** can include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field-programmable gate array (FPGA), logic device, one or more central processing units (CPUs), graphics processing units (GPUs) (e.g., dedicated to efficiently rendering images), processing units performing other specialized calculations, etc. The memory device(s) **84C** can include one or more non-transitory computer-readable storage medium(s), such as RAM, ROM, EEPROM, EPROM, flash memory devices, magnetic disks, etc., and/or combinations thereof.

(158) The memory device(s) **84C** can include one or more computer-readable media and can store information accessible by the one or more processor(s) **84B**, including instructions **84D** that can be executed by the one or more processor(s) **84B**. The instructions **84D** may include one or more steps of

the methods **200**, **400** described above, such as to execute operations at the additive manufacturing apparatus **10** described herein. The memory device(s) **84C** can store instructions **84D** for running one or more software applications, displaying a user interface, receiving user input, processing user input, etc. In some implementations, the instructions **84D** can be executed by the one or more processor(s) **84B** to cause the one or more processor(s) **84B** to perform operations, e.g., such as one or more portions of the methods **200**, **400** described herein. The instructions **84D** can be software written in any suitable programming language or can be implemented in hardware. Additionally, and/or alternatively, the instructions **84D** can be executed in logically and/or virtually separate threads on the processor(s) **84B**.

(159) The one or more memory device(s) **84C** can also store data **84E** that can be retrieved, manipulated, created, or stored by the one or more processor(s) **84B**. The data **84E** can include, for instance, data to facilitate the performance of the methods **200**, **400** described herein. The data **84E** can be stored in one or more database(s). The one or more database(s) can be connected to computing system **84** by a high bandwidth LAN or WAN, or can also be connected to the computing system **84** through the network(s). The one or more database(s) can be split up so that they are located in multiple locales. In some implementations, the data **84E** can be received from another device.

(160) The computing device(s) **84A** can also include a communication module or interface **84F** used to communicate with one or more another component (s) of the computing system **84** or the additive manufacturing apparatus **10** over the network(s). The communication interface **84F** can include any suitable components for interfacing with one or more network(s), including, for example, transmitters, receivers, ports, controllers, antennas, or other suitable components.

(161) It should be appreciated that the additive manufacturing apparatus is described herein only for the purpose of explaining aspects of the present subject matter. In other example embodiments, the additive manufacturing apparatus may have any other suitable configuration and may use any other suitable additive manufacturing technology. Further, the additive manufacturing apparatus and processes or methods **200**, **400** described herein may be used for forming components using any suitable material. For example, the material may be plastic, metal, concrete, ceramic, polymer, epoxy, photopolymer resin, or any other suitable material that may be embodied in a layer of slurry, resin, or any other suitable form of sheet material having any suitable consistency, viscosity, or material properties. For example, according to various embodiments of the present subject matter, the additively manufactured components described herein may be formed in part, in whole, or in some combination of materials including but not limited to pure metals, nickel alloys, chrome alloys, titanium, titanium alloys, magnesium, magnesium alloys, aluminum, aluminum alloys, iron, iron alloys, stainless steel, and nickel or cobalt based superalloys (e.g., those available under the name Inconel® available from Special Metals Corporation). These materials are examples of materials suitable for use in the additive manufacturing processes described herein and may be generally referred to as “additive materials.”

(162) Aspects of the invention(s) are provided by the subject matter of the following clauses, which are intended to cover all suitable combinations unless dictated otherwise based on logic or the context of the clauses and/or associated figures and description:

(163) An additive manufacturing apparatus comprising: a resin support configured to support a first resin and a second resin; a support plate including a window; a stage configured to hold one or more cured layers of the first resin or the second resin to form a component positioned opposite the support plate; a radiant energy device positioned on an opposite side of the resin support from the stage and operable to generate and project radiant energy in a patterned image through the window; and an actuator assembly configured to move the stage in a Z-axis direction and in a Y-axis direction.

(164) The additive manufacturing apparatus of one or more of these clauses, wherein the first resin is laterally offset from the second resin in the Y-axis direction.

(165) The additive manufacturing apparatus of one or more of these clauses, wherein a gap is defined between the first resin and the second resin.

(166) The additive manufacturing apparatus of one or more of these clauses, wherein the resin support comprises a first resin support configured to have the first resin deposited thereon and a second resin support configured to have the second resin deposited thereon.

(167) The additive manufacturing apparatus of one or more of these clauses, wherein the actuator

assembly is further configured to move the radiant energy device along the Y-axis direction relative to the resin support.

(168) The additive manufacturing apparatus of one or more of these clauses, further comprising: a material depositor configured to deposit the first resin and the second resin on the resin support, the material depositor including a first reservoir for retaining the first resin and a second reservoir for retaining the second resin.

(169) The additive manufacturing apparatus of one or more of these clauses, further comprising: a frame operably coupled with the stage, the radiant energy device, and the support plate, the frame further coupled with a mounting plate; and a first slide assembly, wherein the frame, the stage, the radiant energy device, and the support plate are moveable relative to the mounting plate along the first slide assembly.

(170) The additive manufacturing apparatus of one or more of these clauses, wherein an actuator of the actuator assembly is operably coupled with the frame and the mounting plate, and wherein the actuator is configured to move the frame between a first position and a second position along the first slide assembly.

(171) The additive manufacturing apparatus of one or more of these clauses, further comprising: a second slide assembly operably coupled with the radiant energy device and the frame, wherein the radiant energy device is movable relative to the frame along the second slide assembly.

(172) The additive manufacturing apparatus of one or more of these clauses, further comprising: a third actuator operably coupled with the frame and the radiant energy device, wherein the third actuator is configured to move the radiant energy device between a first position and a second position along the second slide assembly.

(173) The additive manufacturing apparatus of one or more of these clauses, further comprising: a frame operably coupled with the stage, the radiant energy device, and the actuator assembly, wherein the frame, the stage, and the radiant energy device are moveable relative to the support plate along a first slide assembly.

(174) A method of operating an additive manufacturing apparatus, the method comprising: depositing a layer of a first resin and a second resin onto a resin support; translating the resin support in an X-axis direction; placing a stage in a first curing position such that a working surface contacts the first resin; curing a portion of the first resin while the stage is in the first curing position; placing the stage in a second curing position, wherein the second curing position is offset from the first curing position in a Y-axis direction; and curing a portion of the second resin while the stage is in the second curing position.

(175) The method of one or more of these clauses, wherein curing the portion of the first resin while the stage is in the first curing position forms a first layer of a component, and curing the portion of the second resin while the stage is in the second curing position forms a second layer of the component.

(176) The method of one or more of these clauses, wherein curing the portion of the first resin while the stage is in the first curing position forms a first portion of a layer of a component and curing the portion of the second resin while the stage is in the second curing position forms a second portion of the layer of the component.

(177) The method of one or more of these clauses, further comprising: positioning a radiant energy device in a first projection position, wherein the portion of the first resin is cured while the radiant energy device is in the first projection position.

(178) The method of one or more of these clauses, further comprising: positioning the radiant energy device in a second projection position, wherein the second projection curing position is offset from the first projection position in the Y-axis direction, and wherein the portion of the first resin is cured while the radiant energy device is in the second projection position.

(179) An additive manufacturing apparatus comprising: a frame including a frame structure, the frame structure having a build plate, a support plate, and a base plate each coupled thereto, wherein the support plate includes a window therein; a first actuator operably coupled with the build plate; a print head operably with the first actuator, wherein the actuator is configured to move the print head relative to a Z-axis direction; a stage operably coupled with the print head; a radiant energy device operably coupled with the frame and positioned on an opposing side of the window from the stage in the Z-axis

direction; and a first slide assembly operably coupled with the base plate of the frame and a mounting plate, the first slide assembly configured to guide movement of the frame relative to the mounting plate in a Y-axis direction.

(180) The additive manufacturing apparatus of one or more of these clauses, wherein the first slide assembly includes a track operably coupled with the mounting plate and one or more guides operably coupled with the base plate, the one or more guides slidable along the track.

(181) The additive manufacturing apparatus of one or more of these clauses, further comprising: a second slide assembly positioned between the radiant energy device and the frame structure, the second slide assembly configured to guide movement of the radiant energy device relative to the window.

(182) The additive manufacturing apparatus of one or more of these clauses, wherein the second slide assembly includes a rail operably coupled with the frame structure and one or more guides, the one or more guides slidable along the rail.

(183) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. An additive manufacturing apparatus comprising: a common resin support configured to support a first resin and a second resin, the first resin laterally offset from the second resin by a gap defined between the first resin and the second resin in a Y-axis direction; a support plate including a window; a stage configured to hold one or more cured layers of the first resin or the second resin to form a component positioned opposite the support plate; a radiant energy device positioned on an opposite side of the common resin support from the stage and operable to generate and project radiant energy in a patterned image through the window; an actuator assembly configured to move the stage in a Z-axis direction and in the Y-axis direction; a frame operably coupled with the stage, the radiant energy device, and the support plate, the frame further coupled with a mounting plate; and a first slide assembly, wherein the frame, the stage, the radiant energy device, and the support plate are moveable relative to the mounting plate along the first slide assembly.

2. The additive manufacturing apparatus of claim 1, wherein an actuator of the actuator assembly is operably coupled with the frame and the mounting plate, and wherein the actuator is configured to move the frame between a first position and a second position along the first slide assembly.

3. The additive manufacturing apparatus of claim 1, further comprising: a second slide assembly operably coupled with the radiant energy device and the frame, wherein the radiant energy device is movable relative to the frame along the second slide assembly.

4. The additive manufacturing apparatus of claim 3, further comprising: a third actuator operably coupled with the frame and the radiant energy device, wherein the third actuator is configured to move the radiant energy device between a first position and a second position along the second slide assembly.

5. An additive manufacturing apparatus comprising: a resin support configured to support a first resin and a second resin; a support plate including a window; a stage configured to hold one or more cured layers of the first resin or the second resin to form a component positioned opposite the support plate, wherein a working surface is defined by one of a surface of the stage or a surface of the component, and wherein the working surface is configured to contact the first resin and the second resin simultaneously; a radiant energy device positioned on an opposite side of the resin support from the stage and operable to generate and project radiant energy in a patterned image through the window; an actuator assembly configured to move the stage in a Z-axis direction and in a Y-axis direction; and a frame operably coupled with the stage, the radiant energy device, and the actuator assembly, wherein the frame, the

stage, and the radiant energy device are moveable relative to the support plate along a first slide assembly.

6. An additive manufacturing apparatus comprising: a frame including a frame structure, the frame structure having a build plate, a support plate, and a base plate each coupled thereto, wherein the support plate includes a window therein; a first actuator operably coupled with the build plate; a print head operably coupled with the first actuator, wherein the first actuator is configured to move the print head relative to a Z-axis direction; a stage operably coupled with the print head; a radiant energy device operably coupled with the frame and positioned on an opposing side of the window from the stage in the Z-axis direction; and a first slide assembly operably coupled with the base plate of the frame and a mounting plate, the first slide assembly configured to guide movement of the frame relative to the mounting plate in a Y-axis direction between a first curing position and a second curing position, wherein the window is configured to be aligned with a first portion of a resin support in the first curing position and a second portion of the resin support in the second curing position, the first portion offset from the second portion in the Y-axis direction.

7. The additive manufacturing apparatus of claim 6, wherein the first slide assembly includes a track operably coupled with the mounting plate and one or more guides operably coupled with the base plate, the one or more guides slidable along the track.

8. The additive manufacturing apparatus of claim 6, further comprising: a second slide assembly positioned between the radiant energy device and the frame structure, the second slide assembly configured to guide movement of the radiant energy device relative to the window.

9. The additive manufacturing apparatus of claim 8, wherein the second slide assembly includes a rail operably coupled with the frame structure and one or more guides, the one or more guides slidable along the rail.

10. The additive manufacturing apparatus of claim 5, wherein an actuator of the actuator assembly is operably coupled with the frame and a mounting plate, and wherein the actuator is configured to move the frame between a first position and a second position along the first slide assembly.

11. The additive manufacturing apparatus of claim 5, further comprising: a second slide assembly operably coupled with the radiant energy device and the frame, wherein the radiant energy device is movable relative to the frame along the second slide assembly.

12. The additive manufacturing apparatus of claim 11, further comprising: a third actuator operably coupled with the frame and the radiant energy device, wherein the third actuator is configured to move the radiant energy device between a first position and a second position along the second slide assembly.

13. The additive manufacturing apparatus of claim 5, further comprising: one or more sensors configured to detect information related to a position of the frame.

14. The additive manufacturing apparatus of claim 5, further comprising: one or more sensors configured to detect information related to a position of the radiant energy device.

15. The additive manufacturing apparatus of claim 1, further comprising: one or more sensors configured to detect information related to a position of the frame.

16. The additive manufacturing apparatus of claim 1, further comprising: one or more sensors configured to detect information related to a position of the radiant energy device.
